



# **Preparing for the Age of AI – A Living Outlook for Decision-Makers**

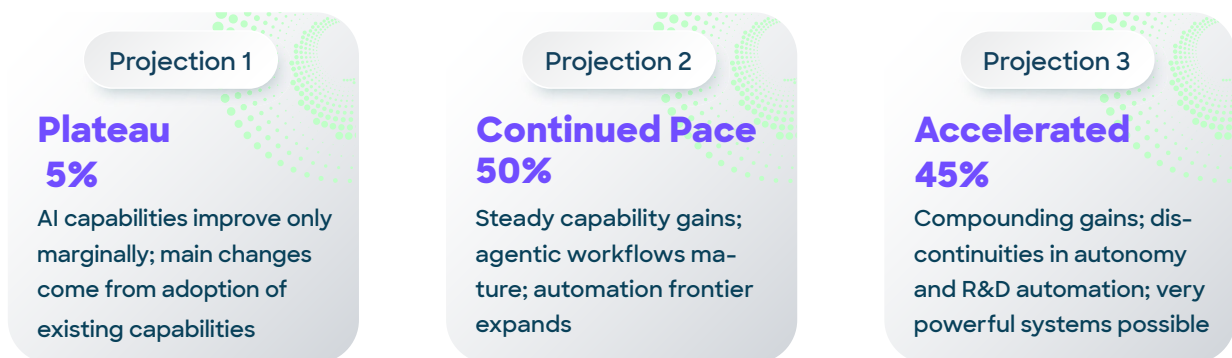
Technological Trajectories in AI: What's Next

# Executive summary

## What this document provides

This AI Perspective Whitepaper provides European decision-makers with a structured orientation for navigating AI advancement over the next 36 months. This report does not present full scenarios. Instead, it offers structured projections of technological capability development as one central driver of future outcomes, providing a framework for preparing across plausible futures.

## Our three perspectives (differentiated by capability speed)



## Key drivers we monitor

1. **Architectures & Training Paradigms** – whether scaling continues, new paradigms emerge (Nested Learning, Causal AI), or both
2. **Agentic Autonomy & Orchestration** – reliability of multi-step autonomous operation (SWE-bench Verified now at 79%, OSWorld at 73%)
3. **Automation of AI R&D and Software Engineering** – feedback loops now materialising (Claude Code writes ~90% of Anthropic's code)
4. **Robotics & Embodied AI** – sim-to-real gap largely closed; Chinese manufacturers achieving 99%+ success rates in production

## Top impacts requiring attention

1. **Labour market pressure** – entry-level knowledge workers already affected (6–20% employment decline in exposed roles)
2. **Information integrity crisis** – 52% of internet content now machine-generated; deepfakes doubling every 6 months
3. **Competitiveness gap** – EU controls only 4.8–5% of global high-end AI compute; 13.48% enterprise adoption vs. 41% for large firms
4. **Institutional adaptation** – only 6% of local governments prioritise AI; 77% of citizens distrust government AI use

## Measures framework

- 19 no-regret measures applicable across all projections (probability = 1)
- 12 projection-conditional measures for P2/P3 preparation
- 5 appliedAI opportunity fields where we focus our contribution

## How to use this document

- **Policy leaders:** Focus on no-regret measures and institutional capacity building
- **Company leaders:** Assess exposure across impact categories; build adoption capacity
- **Ecosystem partners:** Identify collaboration opportunities in measures inventory

## Give us your feedback

We welcome inputs across all aspects of this document:

- **Projections (Chapter 1):** Perspectives on projection framing, baseline conditions, and projection descriptions
- **Drivers & probabilities (Chapter 2):** Insights on main drivers, important trends, and probability assessments
- **Impacts (Chapter 3):** Observations on implications across impact fields; additional evidence or perspectives
- **Measures (Chapter 4):** Additions, ideas, and comments on proposed measures
- **Collaboration (Chapter 5):** If you would like to collaborate with us on our activities or if you want us to highlight your activities

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# Chapter 1: AI advancement projections (3-Year Horizon)

## 1.1 Purpose and scope of this chapter

This chapter presents three plausible projections for AI capability advancement over the next 36 months. The projections are distinguished solely by the speed of capability progress—not by the direction of societal response, regulatory choices, or adoption patterns. Those factors shape outcomes within each projection but do not define the projections themselves.

### Why a 3-year horizon?

Beyond 36 months, uncertainty compounds to the point where projection planning loses decision-relevance. Feedback loops between capability gains, investment cycles, talent flows, and institutional responses become difficult to disentangle. We therefore constrain our outlook explicitly and recommend revisiting assumptions annually.

### What this chapter does not do:

- It does not assign probabilities to projections (see Chapter 2).
- It does not identify the technical drivers that determine which projection materializes (see Chapter 2).
- It does not describe impacts on specific sectors or institutions (see Chapter 3).

Instead, Chapter 1 provides a decision-relevant framing: what each speed regime would mean in practical terms, and what baseline conditions constrain all projections equally.

## 1.2 Key definitions (Working Terms)

To ensure clarity across the document, we define the following terms as used in this whitepaper:

| Term                | Working Definition                                                                                                                                                                                                              |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI advancement      | Measurable increases in the capability and reliability of AI systems on economically and socially relevant tasks. This includes improvements in reasoning, planning, tool use, multimodal understanding, and agentic execution. |
| Agentic AI / Agents | AI systems that plan and execute multi-step tasks using tools, interfaces, and memory, operating under varying degrees of human supervision. Agents may coordinate with other agents or systems to accomplish goals.            |
| Frontier models     | State-of-the-art general-purpose AI models and the products built on them. These systems typically define the current capability ceiling for a given task category.                                                             |
| Automation frontier | The boundary of tasks that can be reliably performed by AI systems with acceptable quality, cost, and oversight. This frontier shifts as capabilities improve.                                                                  |
| Capability speed    | The rate at which the automation frontier expands and at which reliability on complex tasks improves. This is the single dimension that distinguishes our three projections.                                                    |

## 1.3 Baseline conditions: constraints that shape all projections

Before describing the projections, we establish the baseline conditions that apply regardless of which projection materializes. These are structural realities that constrain the speed of change, the feasibility of responses, and Europe’s room for manoeuvre.

### 1.3.1 Compute, energy, and infrastructure cycles

#### Current reality:

AI capability advancement, particularly for frontier models, depends on access to large-scale compute infrastructure. Training runs for leading models require tens of thousands of specialised accelerators (GPUs, TPUs, or custom ASICs) operating in purpose-built data centres with substantial power and cooling requirements. The top 4 US hyperscalers together are investing more than \$600 Billion in 2026 alone, more than the entire German federal budget! [1]

Constraint dynamics within 36 months:

- Data centre buildout is a multi-year process. Site acquisition, permitting, construction, and equipment installation typically require 2–4 years from decision to operation. Projects announced today will largely come online towards the end of our projection horizon or beyond.
- Grid capacity and energy supply are binding constraints in many regions. Even where capital is available, connecting large facilities to reliable power (especially low-carbon power) faces infrastructure and regulatory bottlenecks.
- Chip supply chains remain concentrated. Advanced semiconductor fabrication is dominated by a small number of facilities, primarily outside Europe. Lead times for new fabrication capacity are measured in years.
- Quantum computing will not materially affect AI capabilities within this timeframe. While quantum computing research continues to advance, practical quantum systems capable of accelerating AI training or inference remain years to decades away.

Practical implication:

Within our 36-month window, the physical infrastructure supporting AI advancement will be shaped primarily by decisions already made in unprecedented amounts. New announcements will matter more for the period after our horizon. In the near term, progress will depend heavily on efficiency gains: better algorithms, improved training recipes, optimised inference, and more effective use of existing hardware.

1.3.2 Europe’s dependency exposure

Current reality:

Europe occupies a structurally dependent position in several layers of the AI stack:

| Layer                       | Dependency pattern                                                                                                                                                                                                                          |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chips</b>                | Advanced logic chips (sub-7nm) are fabricated almost exclusively outside Europe (and the US). European chip design capabilities exist but rely on non-European foundries.                                                                   |
| <b>Cloud infrastructure</b> | Hyperscale cloud providers are predominantly US-based. European “alternatives” exist theoretically but lack comparable scale, geographic reach, and ecosystem integration. There is no single player properly suited for large scale AI use |
| <b>Foundation models</b>    | The majority of frontier general-purpose models are developed by US-based (and increasingly Chinese) organisations. European model development is growing but remains significantly behind the capability frontier.                         |
| <b>Talent</b>               | Europe produces strong AI research talent but faces competition for retention, particularly at the applied/engineering end of the spectrum.                                                                                                 |

#### **Constraint dynamics within 36 months:**

- Structural dependencies will not be resolved within this timeframe; building semiconductor fabs or hyperscale infrastructure requires longer cycles.
- However, Europe's ability to deploy, govern, and adapt AI systems is not fully determined by supply-side dependencies. Competence in integration, responsible deployment, and institutional readiness are within European control.

#### **Practical implication:**

Europe's strategic position over the next 36 months will be shaped less by whether it resolves structural dependencies and more by whether it builds operational competence: the ability to adopt AI effectively, govern it responsibly, and maintain agency over critical systems despite upstream dependencies.

### **1.3.3 Institutional and organisational capacity**

#### **Current reality:**

The speed at which AI capabilities translate into real-world impact depends on institutional capacity to adopt, integrate, and govern these systems. This capacity varies widely:

- Public sector: Procurement cycles are slow (often 12–24 months for significant IT projects). Legal and compliance capacity is stretched. IT modernisation backlogs are substantial. Workforce AI literacy is low in most agencies.
- Large enterprises: Adoption is accelerating but unevenly. Many organisations are in pilot or early deployment phases. Integration with legacy systems, data governance, and change management remain significant barriers. [2] [150]
- SMEs: Awareness is growing, but capacity to adopt is constrained by resources, expertise, and access to tailored solutions.

#### **Constraint dynamics within 36 months:**

- Institutional capacity does not scale quickly. Training programmes, procurement reform, and organisational change take time.
- Bottlenecks are not primarily technical; they are organisational, legal, and cultural.
- Even in an “accelerated” capability projection, adoption will be gated by how fast institutions can adapt.

#### **Practical implication:**

In all projections, the execution gap, the difference between what AI systems can do in principle and what organisations actually achieve with them, will be a central determinant of outcomes. Building institutional capacity is a no-regret priority regardless of which projection materializes.

### 1.3.4 Competitive pressure and forced adaptation

#### Current reality:

While institutional capacity constrains adaptation speed, competitive pressure can override organisational inertia. In globally competitive markets, the pace of adoption is increasingly set not by internal readiness but by external necessity.

#### Emerging dynamics:

- Speed-to-market compression: Organisations that integrate AI into product development and operational workflows can dramatically reduce cycle times. When a competitor releases software updates in weeks rather than months, or responds to customer requests with high-quality proposals in minutes rather than days, the market benchmark shifts for everyone.
- Productivity arbitrage: Firms that achieve higher output per employee through AI augmentation gain structural cost and quality advantages. Competitors that cannot match this productivity face margin pressure, talent drain, and market share loss.
- Winner-take-more dynamics: In markets where speed and quality advantages compound (e.g., software, professional services, design, engineering), early adopters may pull ahead faster than laggards can catch up.

#### The “San Francisco consensus”:

In July 2025, Eric Schmidt, former Google CEO, described a belief among Silicon Valley technology leaders that “in the next two to four years (the average is three years) the entire world will change.” This expectation of rapid, transformative change is driving investment decisions, talent allocation, and competitive behaviour in ways that affect European organisations regardless of their own beliefs about AI timelines. [3]

#### Constraint dynamics within 36 months:

- For organisations operating in globally competitive, exposed sectors, adaptation may not be optional. The choice becomes: adopt rapidly or face existential competitive risk.
- For sheltered sectors (parts of the public sector, regulated industries, local services), competitive pressure is less immediate but not absent. Indirect effects, such as talent expectations, citizen comparisons, and supply chain dependencies, will transmit pressure over time.

#### Practical implication:

Institutional readiness is a constraint, but market pressure is a forcing function. In projections with rapid capability gains (P2 and especially P3), organisations in competitive markets will be compelled to adapt faster than their internal change capacity would normally allow—with significant implications for workforce, governance, and organisational stability.

## 1.4 The three projections: differentiated by capability speed

We now describe three projections for AI capability advancement over the next 36 months. Each projection is defined by the speed at which AI systems improve in capability and reliability on economically and socially relevant tasks.

Important: These projections describe capability trajectories, not outcomes. Outcomes depend on how societies, institutions, and organisations respond—which is addressed in Chapters 3 and 4.

### Projection 1 (P1): Plateau

AI capabilities improve only marginally compared to today's frontier models.

What this projection looks like:

In this projection, the rapid capability gains observed in recent years do not continue at the same pace. Progress becomes incremental rather than step-changing. The fundamental architecture and training paradigms that underpin current systems encounter diminishing returns or unforeseen bottlenecks.

#### Typical improvements in this projection:

- Incremental gains in reliability, latency, and cost efficiency
- Better user interfaces and integration tooling
- Modest improvements in specific domains through fine-tuning and specialisation
- Continued reduction in deployment friction (APIs, SDKs, enterprise connectors)
- What remains difficult:
- Robust long-horizon autonomy (multi-step tasks over extended timeframes with minimal supervision)
- Consistently correct reasoning under ambiguity and uncertainty
- Reliable operation in open-ended, high-stakes environments
- Verifiable alignment with complex human intentions
- Transfer of virtually trained models to real-world robotic applications (sim-to-real gap remains a significant barrier, limiting practical deployment of robots in unstructured environments)

#### Why this projection still matters:

While AI projection probabilities are highly contested among experts, P1 should be considered the generally accepted lower boundary for capability advancement. Thus, all impacts described for P1 should be understood as the minimum expected impacts, and all measures identified for P1 should be considered the absolute minimum response required.

A plateau in capability advancement does not mean a plateau in impact. Current AI systems are already capable enough to transform significant portions of knowledge work, administrative processes, and creative production. The question is whether organisations can adopt and integrate these capa-

bilities effectively. [4] [5]

In Projection 1, the primary driver of change is diffusion and adoption, not new capabilities. Organisations that invest in process redesign, workforce development, and systematic integration can realise substantial productivity gains. Those that do not will fall behind—not because of new AI breakthroughs, but because competitors extract more value from existing capabilities.

### **Projection 2 (P2): Continued pace**

AI capabilities continue to improve at roughly the pace observed in recent years; agentic workflows become materially more useful.

#### **What this projection looks like:**

In this projection, the trajectory of recent years continues. Frontier models become meaningfully more capable through a combination of scaling, algorithmic improvements, better training data, and advances in post-training techniques (reinforcement learning, inference-time reasoning, tool use). Progress is steady but not explosive.

#### **Typical improvements in this projection:**

- Stronger planning and multi-step reasoning capabilities
- More reliable tool use and structured outputs (code, data manipulation, API calls)
- Improved grounding on enterprise data and context
- Better multimodal handling (text, image, audio, video in combination)
- Partially autonomous workflows for bounded domains (software development, analytics, customer service, document processing)
- Significant improvement in sim-to-real transfer for robotics, enabling practical deployment of robots in diverse fields of application (logistics, manufacturing, service environments)

#### **New “normal” capabilities within 12–36 months:**

- AI agents that can execute multi-step workflows with moderate supervision (e.g., draft → review → revise cycles in writing, coding, or analysis)
- Autonomous handling of well-defined operational tasks (scheduling, monitoring, routine decisions) under explicit controls
- Meaningful productivity gains in software engineering, back-office operations, and knowledge-intensive services
- Software development becomes substantially automated, with AI handling the majority of routine coding, testing, and debugging tasks
- Core organisational processes begin running with limited human supervision in leading organisations

### Adoption pattern:

Diffusion remains uneven. Organisations with strong change capacity, data infrastructure, and AI-literate workforces pull ahead. Laggards, including many public sector bodies and SMEs, struggle to keep pace. The gap between leaders and followers widens.

### Projection 3 (P3): Accelerated

AI capabilities compound rapidly; discontinuities in autonomy and R&D automation become plausible within 36 months. Some experts predict the emergence of very powerful AI, maybe even AGI (Artificial General Intelligence) systems within this timeframe.

### What this projection looks like:

In this projection, multiple drivers of progress align and reinforce each other. Advances in architectures, training methods, or emergent capabilities produce step-changes that accelerate the rate of improvement itself. AI systems begin to contribute meaningfully to AI research and software engineering, creating feedback loops that compress timelines significantly.

The possibility of reaching highly capable, general-purpose AI systems (sometimes referred to in public discourse as approaching “AGI-level” capabilities) enters the plausible range within 36 months. While definitions vary and uncertainty remains high, the practical implication is that AI systems may achieve or approach human-level performance across a wide range of cognitive tasks, not just narrow domains.

### Expert views on accelerated timelines:

Leading AI researchers and executives have offered increasingly specific predictions about AGI timelines during 2025–2026:

| Expert         | Organisation      | Timeline view      | Key statement                                                                                                      | Source            |
|----------------|-------------------|--------------------|--------------------------------------------------------------------------------------------------------------------|-------------------|
| Dario Amodei   | Anthropic CEO     | 2026–2027          | „A country of geniuses in a data centre“ matching Nobel Prize-level capabilities; 50% probability within 1–3 years | [6] [15]          |
| Sam Altman     | OpenAI CEO        | Mid-2020s          | „We are now confident we know how to build AGI“; superintelligence within 5 years                                  | [7]               |
| Demis Hassabis | DeepMind CEO      | 2028–2030          | Revised from „5–10 years“ to „3–5 years away“; 50% by end of decade                                                | [8] [9] [10] [15] |
| Eric Schmidt   | Former Google CEO | 2–6 years (avg. 3) | „San Francisco Consensus“ among tech leaders; recursive self-improvement within a decade                           | [3]               |

|                |                                                  |                                              |                                                                                                 |           |
|----------------|--------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------|-----------|
| Shane Legg     | DeepMind Co-founder                              | 2028 (minimal AGI)                           | 50% probability of minimal AGI by 2028; full AGI 3–6 years later                                | [11]      |
| Ilya Sutskever | SSI CEO                                          | Research-dependent                           | 2025 onward as return to „age of research“ where scaling alone insufficient                     | [12]      |
| Yoshua Bengio  | AI Pioneer                                       | Few years to decade                          | „Many researchers now consider human-level AI plausible within a few years to a decade“         | [13] [15] |
| Yann LeCun     | Former Meta Chief Scientist (left December 2025) | Not within 2 years, but „probably very soon“ | Argues current LLM approaches alone insufficient; pursuing world models at AMI Labs             | [14] [15] |
| Llion Jones    | Sakana.ai Co-founder                             | Focused on recursive self-improvement        | Estimates 1–2 breakthroughs needed for AGI; believes recursive self-improvement is the key path | [15]      |

### The architecture debate:

A fundamental disagreement exists among experts about whether AGI requires new architectures beyond transformers: [

- Scaling advocates (Altman, Amodei): Current transformer paradigm with refinements (reasoning modules, tool use, inference-time compute) is likely sufficient. OpenAI's o-series models achieving 75.7% on ARC-AGI (vs. GPT-4o's 5%) suggest transformers with reasoning can adapt to novel tasks. [16]
- Architecture innovators (LeCun, Jones, Markram): Transformers face fundamental limitations. Yann LeCun left Meta in December 2025 to launch AMI Labs (€3B valuation) focused on «world models» rather than scaling LLMs. Llion Jones co-founded Sakana.ai to focus on recursive self-improvement, estimating 1–2 key breakthroughs are needed. Henry Markram's INAIT is pursuing causal learning as the foundation for understanding AI rather than pattern-matching AI. [14] [17] [18]
- Hybrid perspective (Hassabis, Bengio): Additional breakthroughs needed, particularly continual learning, world models, reasoning, hierarchical planning, and creative hypothesis generation, but these may build on rather than replace current paradigms. [19]

### Typical improvements in this projection:

- Rapid cycles of capability increases, with shorter intervals between major advances
- Substantially more autonomous execution across broad task categories
- Meaningful AI contribution to research and engineering workflows (experiment design, code generation, debugging, optimisation, scientific discovery)
- Emergence of capabilities that were not explicitly trained or anticipated

- Robotic applications expand dramatically, including viable deployment of humanoid robots in commercial and industrial settings, enabled by breakthroughs in sim-to-real transfer, perception, and manipulation

#### **Agentic capabilities:**

- Autonomy in substantial end-to-end processes becomes viable, not just bounded subtasks
- AI systems can manage complex, multi-stage projects with minimal human intervention
- Coordination between multiple AI agents becomes routine for complex workflows
- Human oversight shifts from task-level supervision to goal-level governance

#### **Stress points:**

An accelerated projection creates significant stress across institutions and systems:

- Policy and regulation: Governance frameworks designed for slower change may become outdated before implementation is complete. The gap between capability and governance widens.
- Education and workforce: Curricula and training programmes cannot adapt quickly enough; cohorts of workers face rapid skill obsolescence across many professions simultaneously.
- Public services: Demand for support (unemployment, retraining, fraud prevention, social stability) may spike while institutional capacity lags severely.
- Information integrity: The scale and sophistication of synthetic content (text, image, audio, video) may fundamentally outpace detection and attribution capabilities, challenging the foundations of trust in digital communication.
- Economic structure: Fundamental questions about value distribution, employment, and economic organisation may arise faster than political systems can address them.

#### **Competitive dynamics and forced adoption:**

In this projection, competitive pressure becomes extreme. Organisations in globally competitive markets face existential pressure to adopt AI at a pace that may exceed their capacity for orderly change:

- Product and service cycles compress dramatically; organisations that cannot keep pace lose market position within months, not years.
- The productivity gap between AI-native organisations and traditional competitors becomes a chasm.
- Workforce transformation cannot wait for gradual reskilling; organisations must simultaneously deploy AI and manage significant workforce restructuring.
- Some organisations will thrive by embracing radical transformation; others will face severe disruption or failure despite best efforts, simply due to the speed of change.

## 1.5 Projection comparison: one-page decision aid

The following table summarises the three projections across key dimensions. It is intended as a quick reference for decision-makers, not a comprehensive analysis.

| Dimension              | P1: Plateau                                            | P2: Continued Pace                                     | P3: Accelerated                                                                                    |
|------------------------|--------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Capability speed       | Marginal improvements; gains mostly in cost/efficiency | Steady capability gains; regular new model generations | Compounding gains; potential discontinuities; very powerful systems possible                       |
| Automation frontier    | Broadly similar to today; expands via adoption         | Expands into multi-step workflows and bounded autonomy | Expands rapidly; autonomy in substantial end-to-end processes                                      |
| Reliability trajectory | Improves slowly; edge cases remain problematic         | Improves materially; deployment confidence grows       | Uneven but fast; reliability sufficient for autonomous operation in many domains                   |
| Agentic viability      | Limited; supervision-heavy                             | Growing; practical for bounded enterprise tasks        | Autonomy in substantial end-to-end processes; goal-level oversight replaces task-level supervision |
| Robotics               | Sim-to-real gap persists; limited practical deployment | Diverse robotic applications become viable             | Broad deployment including humanoids; robotics becomes economically transformative                 |
| Main bottleneck        | Adoption and integration capacity                      | Adoption + governance + workforce readiness            | Societal and institutional adaptation speed; governance gap                                        |
| Diffusion pattern      | Leaders gain from existing capabilities                | Leaders widen gap over laggards                        | Extreme divergence; forced rapid adoption in exposed sectors; potential for systemic shocks        |
| Europe's primary risk  | Complacency and slow diffusion                         | Widening competitiveness gap                           | Loss of control/agency; dependency deepens; institutional overload                                 |
| Urgency of response    | Moderate (but not negligible)                          | High                                                   | Very high                                                                                          |

## 1.6 What this chapter does not determine

To avoid misinterpretation, we explicitly note what this chapter leaves open:

1. Projection probabilities: We do not assign likelihoods in this chapter. Probability estimates require analysing the underlying drivers (Chapter 2).
2. Which projection is “good” or “bad”: Each projection contains both risks and opportunities. Outcomes depend on responses, not just capability trajectories.
3. Specific impacts: How each projection affects labour markets, public services, democracy, or competitiveness is addressed in Chapter 3.
4. What to do: Measures and priorities are derived in Chapter 4, after impacts are assessed.
5. Technological drivers: The technical factors that determine which projection materializes—architectures, agentic reliability, R&D automation, robotics—are analysed in Chapter 2.

## 1.7 How to read the rest of this document

With the projection frame established, the remainder of this whitepaper proceeds as follows:

- Chapter 2 identifies the key technical drivers that determine projection probabilities and provides our current best estimates.
- Chapter 3 maps concrete impacts (opportunities and risks) across sectors and systems for each projection.
- Chapter 4 derives measures from those impacts, distinguishes no-regret actions from projection-conditional ones, and specifies the appliedAI Institute's portfolio response.
- Chapter 5 describes how we keep this document current: continuous monitoring, expert input, and versioning protocols.

The goal is a decision-ready orientation: not a prediction of what will happen, but a structured basis for preparing across plausible futures.

# Chapter 2:

## Speed drivers & projection probabilities

### 2.1 Purpose of this chapter

Chapter 1 established three projections differentiated solely by the speed of AI capability advancement. This chapter identifies the technical drivers that determine which projection materializes and provides our current best-estimate probability assessments.

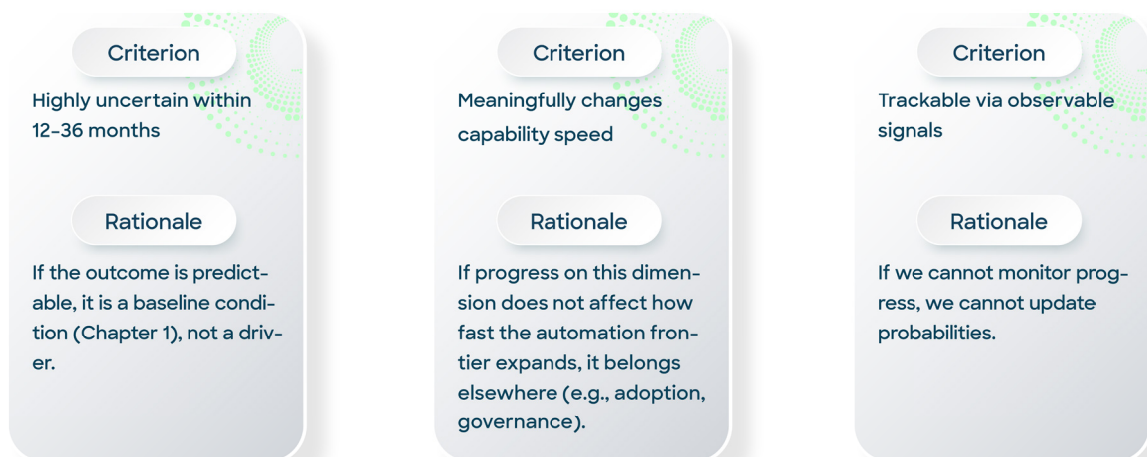
The logic:

1. Identify a small set of drivers that are (a) highly uncertain within 36 months, (b) meaningfully affect capability speed, and (c) can be tracked via observable signals.
2. For each driver, describe what P1/P2/P3 would look like for that driver specifically.
3. Consolidate driver assessments into projection probabilities.
4. Specify breakthrough triggers that would prompt re-estimation.

Note on probability estimates: We publish point estimates as a decision aid, not as precise forecasts. These estimates reflect our current best judgment informed by available evidence. Probabilities are re-estimated when monitoring detects material breakthroughs or disruptions.

### 2.2 Driver selection criteria

A topic belongs in this chapter only if it meets all three criteria:



### What is excluded from this chapter:

- Compute/energy buildout: Largely determined by decisions already made; affects all projections similarly within 36 months. (Covered in Chapter 1 as a baseline condition.)
- Open-source ecosystem health: Affects Europe's ability to respond, not capability speed itself. (Relevant to Chapter 4.)
- Regulation and standards: Affects deployment and governance, not fundamental capability advancement. (Relevant to Chapters 3 and 4.)
- Security and misuse: These are impacts of capability advancement, not drivers of it. (Covered in Chapter 3.)

## 2.3 The four core drivers

Based on the selection criteria, we identify four drivers that will determine which projection materializes:

1. Architectures & Training Paradigms
2. Agentic Autonomy & Orchestration
3. Automation of AI R&D and Software Engineering
4. Robotics & Embodied AI

Each driver is analysed below with:

- Definition and mechanism
- What P1/P2/P3 looks like for this driver
- Current evidence and expert views
- Signals and indicators to monitor
- Directional impact on projection probabilities

## 2.4 Driver 1: Architectures & training paradigms

### 2.4.1 What this driver means

This driver captures whether AI capability gains continue to come primarily from scaling and engineering improvements within current paradigms (primarily transformer-based architectures), or whether new architectural approaches emerge that unlock qualitatively different capabilities.

#### Key questions:

- Do we see continued gains from scaling (more compute, more data, larger models)?
- Do post-training techniques (RLHF, inference-time reasoning, chain-of-thought) continue to yield meaningful improvements?
- Does a genuinely new paradigm emerge (e.g., world models, hybrid neuro-symbolic systems, nested learning, causal AI, or fundamentally different architectures)?

#### Emerging paradigms: nested learning and causal ai

Two significant architectural innovations have emerged in 2025–2026 that may represent pathways beyond transformer limitations:

##### Google DeepMind's Nested Learning (NeurIPS 2025):

Nested Learning, introduced in the paper “Nested Learning: The Illusion of Deep Learning Architectures,” bridges the gap between model architecture and optimisation algorithms by treating them as a unified system of interconnected, multi-level optimisation problems. Key innovations include:

- Continuum Memory Systems (CMS): Memory is viewed as a spectrum of modules, each updating at different frequency rates, creating richer memory systems for continual learning—directly addressing the «catastrophic forgetting» problem that limits current LLMs.
- Deep Optimizers: By viewing optimizers as associative memory modules, the approach derives new formulations for concepts like momentum that are more resilient to imperfect data.
- Hope Architecture: A proof-of-concept self-modifying architecture that achieves superior performance in language modelling and demonstrates better long-context memory management than existing state-of-the-art models including Titans, TTT, and Mamba2.

The Hope architecture demonstrates that principled unification of architecture and optimisation can lead to more expressive, capable, and efficient learning algorithms—potentially closing the gap between current LLM limitations and human-like continual learning abilities. [20] [21]

##### EPFL/INAIT Causal AI:

Henry Markram, who led the Blue Brain Project and Human Brain Project, has launched INAIT to commercialise causal learning—a fundamentally different approach to AI that focuses on understanding cause and effect rather than pattern correlation:

- Causal vs. Correlational Learning: Current AI observes patterns; INAIT's approach enables AI to participate in and understand the causal dynamics of systems, a prerequisite for genuine understanding and autonomous action.
- Three Product Lines: Future Complete (causal forecasting), Cognition Complete (causal decision-making), and Behaviour Complete (full-scale robotics with continuous learning).
- Transparency Advantage: Causal AI provides inherent explainability because decision chains can be traced through causal relationships, addressing a key limitation of current «black box» neural networks.
- Commercial Deployment: INAIT's Future Complete product launches in 2026, with Microsoft as distribution partner, targeting financial services, supply chain, and robotics applications.

Markram describes this as moving from “knowledge AI” (pattern matching) to “understanding AI” (causal reasoning), a transition that could be as significant as the original deep learning revolution. [18] [22]

## 2.4.2 Expert views on architectures and timelines

Leading AI experts hold divergent views on the role of new architectures in reaching very powerful AI systems:

| Expert         | Organisation                           | AGI Timeline                                 | Architecture View                                                         | Breakthroughs Needed                   | Source       |
|----------------|----------------------------------------|----------------------------------------------|---------------------------------------------------------------------------|----------------------------------------|--------------|
| Dario Amodei   | Anthropic                              | 2026–2027                                    | Current paradigm + scaling + AI-assisted R&D likely sufficient            | Very few; mostly continuation          | [6]          |
| Sam Altman     | OpenAI                                 | Mid-2020s                                    | Transformer-style models extended with tools, agents, compute             | Small number of insights + engineering | [7]          |
| Demis Hassabis | DeepMind                               | 2028–2030                                    | Needs world models, continual learning, planning, robotics–hybrid systems | Several major but related advances     | [8] [9] [10] |
| Yann LeCun     | Former Meta Chief Scientist / AMI Labs | Not within 2 years, but „probably very soon“ | LLMs alone insufficient; needs new architectures with world models        | Multiple breakthroughs; new paradigm   | [14]         |

|               |                      |                                       |                                                                                   |                                      |      |
|---------------|----------------------|---------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------|------|
| Llion Jones   | Sakana.ai Co-founder | Focused on recursive self-improvement | Recursive self-improvement is the key path; current paradigm may be local optimum | 1-2 key breakthroughs needed for AGI | [15] |
| Shane Legg    | DeepMind             | 2028 (minimal)                        | Distinguishes minimal vs. full AGI; scaling may suffice for minimal               | Minimal: few; Full: several more     | [11] |
| Henry Markram | EPFL / INAIT         | Imminent for domain-specific          | Causal learning required for understanding; current AI only correlates            | Causal AI solves key limitation      | [18] |

#### The architecture debate in detail:

##### Transformer advocates argue that the current paradigm, with refinements, can reach AGI:

- OpenAI's o-series models demonstrate that inference-time reasoning can dramatically improve performance on novel tasks (o3 achieved 75.7% on ARC-AGI vs. GPT-4o's 5%)
- Continued improvements in training recipes, data curation, and post-training techniques suggest substantial headroom remains

##### Architecture innovators argue that fundamental limitations require new approaches:

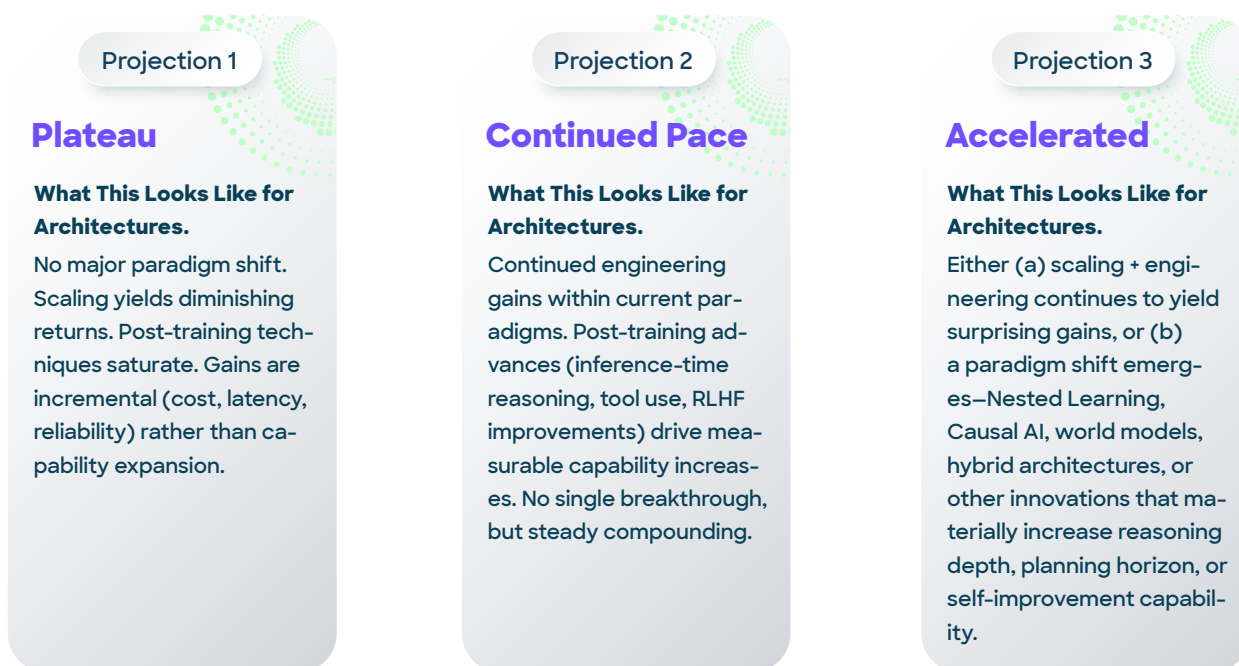
- Yann LeCun stated at Davos 2026 that AGI will not arrive within 2 years but is "probably coming very soon," and that transformers cannot learn causality or understand physical dynamics. He launched AMI Labs (€3B valuation) in December 2025 specifically to pursue "world models."
- Llion Jones (transformer co-inventor) co-founded Sakana.ai to focus on recursive self-improvement, estimating that 1-2 key breakthroughs are needed for AGI. He warns of a "gravitational well" where transformer success prevents exploration of alternatives.
- Henry Markram argues that current AI captures knowledge but doesn't understand it—AI is an "observer" while brains are "participants" in the world. INAIT's causal learning enables AI to understand cause and effect, which he considers prerequisite for true intelligence.
- Missing capabilities identified by multiple experts: continual learning, world models, hierarchical planning, creative hypothesis generation, causal reasoning.

##### Hybrid perspectives suggest breakthroughs within or extending current paradigms:

- Hassabis identifies specific missing capabilities but frames them as extensions within deep learning
- Google's Nested Learning approach demonstrates that continual learning can be achieved through architectural innovation within neural network paradigms

- Joint Embedding Predictive Architecture (JEPA) variants from Meta represent attempts to add world-model capabilities to neural approaches
- Neuro-symbolic integration combining neural networks with structured reasoning

### 2.4.3 Driver-specific projection mapping



### 2.4.4 Signals and indicators to monitor

| Signal                                             | Why It Matters                                                            | What We Track                                                | Current Read                                                                       | Confidence  |
|----------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|-------------|
| ARC-AGI-2 and novel reasoning benchmarks           | Tests genuine adaptation with both high adaptability and high efficiency; | ARC-AGI-2 scores, cost efficiency                            | Human Panel: 100% at \$17; Gemini 3 Deep Think: 84.6% at \$13.62                   | Medium-High |
| New architecture announcements with verified gains | Would signal paradigm shift                                               | Major lab releases, benchmark results, research publications | AMI Labs, Nested Learning, INAIT, Sakana.ai launched; multiple approaches emerging | Medium      |

|                                   |                                                        |                                                                                                               |                                                                                                        |             |
|-----------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------|
| Architectural innovation research | Tests whether new paradigms emerge beyond transformers | Papers on continual learning, causal AI, recursive self-improvement, world models, neuro-symbolic integration | Active research across multiple organisations; growing publication volume                              | Medium-High |
| Compute infrastructure scaling    | Tests continued validity of scaling laws               | Data centre capacity announcements, training run scales                                                       | OpenAI: 1 GW capacity early 2026, additional 1 GW by end of year; major scaling investments continuing | High        |
| World model demonstrations        | Core capability gap identified by sceptics             | DeepMind Genie, Meta JEPA, AMI Labs                                                                           | Research active; no production deployment                                                              | Low-Medium  |

## 2.4.5 Directional impact on projection probabilities

- Evidence of paradigm shift (Nested Learning, Causal AI, recursive self-improvement, or other approaches achieving production deployment with verified superior performance): shifts probability towards P3 [23]
- Sustained engineering gains without paradigm shift: supports P2
- Diminishing returns on scaling and post-training: shifts towards P1

Current Assessment: The emergence of multiple promising architectural innovations—including Nested Learning (DeepMind, NeurIPS 2025) demonstrating superior continual learning and memory management, INAI's commercial launch of Causal AI with Microsoft partnership, Richard Socher's work on recursive self-improvement, AMI Labs pursuing world models, and other organisations exploring neuro-symbolic integration—provides strong evidence that architectural innovation is accelerating across the field. These are not isolated efforts but represent a broader trend of multiple companies and research groups pursuing alternatives and extensions to current paradigms.

Furthermore, it is reasonable to assume that architectural innovations will continue to emerge. We observe how efficiently the biological brain operates compared to current AI models—the brain achieves remarkable capabilities with approximately 20 watts of power and learns from vastly less data than current models require. This efficiency gap suggests substantial room for improvement in AI architectures.

Additionally, all scaling laws for pre-training, mid-training, and post-training continue to hold. OpenAI's expansion to 1 GW of data centre capacity in early 2026, with an additional 1 GW planned by year-end, indicates continued confidence in scaling approaches. It is reasonable to assume that with significantly larger compute resources, substantially better models will emerge.

The combination of (a) multiple promising architectural paths reducing P1 probability, (b) strong continued investment in scaling supporting P2, and (c) potential for architectural breakthroughs or scaling surprises supporting P3, creates the current probability distribution where both P2 and P3 have substantial weight.

## 2.5 Driver 2: Agentic autonomy & orchestration

### 2.5.1 What this driver means

This driver captures the practical ability of AI systems to execute multi-step goals using tools, memory, and coordination patterns—with acceptable reliability, cost, and oversight requirements.

**Key questions:**

- Can AI agents reliably complete bounded but meaningful workflows (not just single tasks)?
- How fast is the “time horizon” of reliable autonomous operation expanding?
- Can multiple agents coordinate effectively on complex tasks?
- What supervision level is required for acceptable error rates?

### 2.5.2 Current benchmark evidence (February 2026)

The benchmark landscape for agentic AI has matured substantially, with dramatic improvements across all major benchmarks:

**Real-world task performance:**

| Benchmark          | What It Measures                    | Best Per-for- | Leading Model/System           | Human Baseline   | Gap             | Source |
|--------------------|-------------------------------------|---------------|--------------------------------|------------------|-----------------|--------|
| SWE-bench (bash)   | Real GitHub issue resolution        | 76.8%         | Claude Opus 4.5                | ~70–80%          | Closed/Exceeded | [24]   |
| SWE-bench Verified | Human-verified solvable issues      | 79.2%         | Claude Opus 4.5 (High Compute) | 100% (by design) | 21 points       | [25]   |
| GAIA               | Multi-step research/reasoning tasks | 74.55%        | Claude Sonnet 4.5              | 92%              | 17 points       | [26]   |
| WebArena           | Interactive web task completion     | 74.3%         | DeepSeek v3.2                  | ~85%             | 11 points       | [27]   |
| OSWorld            | Real OS environment tasks           | 73%           | Multimodal agent system        | 72%              | Closed          | [28]   |

### Key observations:

- SWE-bench has reached human parity. The benchmark uses 2,294 real GitHub issues; success requires understanding complex codebases, identifying root causes, and implementing working patches. Claude Opus 4.5 at 79.2% now matches or exceeds estimated human performance.
- OSWorld gap has closed dramatically. The jump from 12.24% to 73% represents one of the most significant capability improvements in the past year, demonstrating that real computer environment interaction—previously the largest gap—is now approaching human-level performance.
- GAIA shows strong progress towards the 92% human baseline, with Claude Sonnet 4.5 achieving 74.55% on multi-step research and reasoning tasks requiring tool use and complex problem-solving.
- WebArena has seen major improvement with DeepSeek v3.2 achieving 74.3%, substantially closing the gap to human performance and demonstrating that web automation is becoming practically viable.

### Time-horizon analysis:

Research suggests frontier agents can now reliably complete tasks equivalent to hours of expert human work for bounded domains, with effective time horizons expanding rapidly. Key benchmarks measuring task duration capability:

#### METR task horizon benchmark:

METR proposes measuring AI performance in terms of the length of tasks AI agents can complete. Their research shows that this metric has been consistently exponentially increasing over the past 6 years, with a doubling time of around 7 months. Extrapolating this trend predicts that, by the end of the 3-year timeframe of this document, we will reach 2 weeks of continuous work done by agents within a fraction of a time. [29]

- Current state of the art: GPT-5.2 achieves 6 hours 34 minutes task length for software tasks completed 50% of the time. [30]

#### Vending-bench 2:

Vending-Bench 2 is an advanced AI benchmark developed by Andon Labs (released late 2025/early 2026) that evaluates the long-term, agentic capabilities of Large Language Models. It simulates a 365-day, high-stakes projection where an AI agent must operate a vending machine business, focusing on maintaining coherence, managing inventory, setting prices, and maximising profit over a long time horizon (>20M tokens).

- Current state of the art: Claude Opus 4.6 achieves approximately \$7,500 profit over the simulated year.
- Progress indicator: Six months ago, frontier models achieved roughly \$0 profit on this benchmark, indicating dramatic improvement in long-horizon coherence and planning.

The SWE-bench results demonstrate that agents can now handle substantial software engineering tasks that previously required experienced developers working over extended periods. Combined with METR's exponential trend and Vending-Bench 2's demonstration of year-long planning coherence, the evidence indicates that agentic time horizons are expanding faster than many anticipated. [31]

## 2.5.3 OpenClaw: The emergence of system-integrated agents (updated)

OpenClaw (formerly Clawdbot or Moltbot) has emerged as one of the most significant frameworks for agentic autonomy, marking the transition from passive chatbots to proactive, system-integrated agents that can act independently. Its emergence surprised the entire AI community with how quickly system-level agent integration became practically viable.

### Core capabilities:

- **System Integration (“AI with Hands”):** Unlike isolated web interfaces, OpenClaw runs locally (macOS, Linux, Windows) with root-level access to the file system, shell commands, and browser control.
- **Proactive Action:** Through «heartbeats» and cron jobs, agents can initiate tasks (e.g., portfolio monitoring, email triage) without direct user prompts.
- **Persistent Memory:** Stores interaction histories and user preferences locally (e.g., as Markdown), enabling adaptive, session-consistent personality.

### Orchestration architecture:

- **Model-Agnostic Routing:** Functions as a gateway that dynamically routes tasks to different LLMs (Claude, GPT, DeepSeek), including automatic failover chains.
- **Declarative Skills System:** Capabilities defined through structured SKILL.md files, enabling controlled and auditable orchestration where agents access only context-relevant tools.
- **Multi-Agent Teams:** Users can configure specialised agent families that delegate tasks and track progress among themselves. [32]

### Security concerns:

Despite its capabilities, OpenClaw is considered an “identity security nightmare” for enterprises:

- **Privilege Escalation:** Agents often inherit full user permissions, which can lead to critical system interventions with unpredictable behaviour.
- **Vulnerability Rate:** Approximately 26% of community-provided «skills» showed security vulnerabilities in analyses.
- **Decentralisation:** Because OpenClaw runs locally and distributed, there is no central «kill switch,» complicating regulatory control.

OpenClaw represents the leading tool for users seeking maximum functional independence for their agents, but requires high technical competence in security and governance. [33]

## Implications for Projection Assessment:

The rapid emergence and adoption of OpenClaw demonstrates that system-integrated agentic AI is not a future possibility but a current reality. This has several implications:

1. Capability validation: The fact that users are successfully deploying agents with file system access, browser control, and proactive task initiation validates that agentic reliability has reached practical thresholds.
2. Security concerns become urgent: The 26% vulnerability rate in community skills indicates that agentic deployment is outpacing security governance—a pattern that will likely intensify.
3. Enterprise adoption pressure: Despite security concerns, competitive pressure will drive enterprise adoption of similar capabilities, forcing organisations to develop governance frameworks rapidly.

## Outlook:

We expect in 2026 that this space will further develop along two tracks:

- Corporate versions: Enterprise-grade variants of OpenClaw with improved security controls, audit capabilities, and centralised governance will emerge from major vendors.
- Malicious use: Bad actors will adopt similar tools for automated fraud, social engineering at scale, and autonomous cyberattack orchestration, requiring defensive capabilities to advance in parallel.

## 2.5.4 Driver-specific projection mapping



## 2.5.5 Signals and Indicators to Monitor

| Signal                                   | Why It Matters                                | What We Track                               | Current Read                                | Confidence |
|------------------------------------------|-----------------------------------------------|---------------------------------------------|---------------------------------------------|------------|
| SWE-bench progression                    | Direct measure of practical coding capability | Quarterly leaderboard updates               | 77.5-82% best; at/above human level         | High       |
| OSWorld progression                      | Real environment interaction                  | Leaderboard updates                         | 73%; dramatic improvement                   | High       |
| GAIA scores                              | Holistic assistant capability                 | Top scores, gap to human                    | 74.55% vs. 92% human                        | High       |
| Enterprise deployment announcements      | Market validation                             | Major vendor releases, case studies         | Growing rapidly; production deployments     | High       |
| OpenClaw adoption and security incidents | System-integrated agent maturity              | Community metrics, incident reports         | High adoption; security concerns documented | Medium     |
| Multi-agent coordination results         | Complex workflow viability                    | Research benchmarks, production deployments | Active development; production emerging     | Medium     |

## 2.5.6 Directional Impact on Projection Probabilities

- Continued benchmark improvement + successful enterprise deployment: strongly supports P2/P3
- OSWorld gap widens: strong signal towards P3
- Major security incidents with agentic systems: could slow deployment but unlikely to reverse capability
- SWE-bench exceeding 90%: would indicate P3 trajectory

Current Assessment: The dramatic improvement in SWE-bench (to 77-80%, matching human performance), OSWorld (from 12% to 73%, matching human performance), and GAIA (to 74.55%) represents transformative progress. Combined with the emergence of system-integrated frameworks like OpenClaw, the evidence strongly supports P2 as the baseline with significant probability of P3. The agentic capability gap that previously distinguished projections has largely closed within the past year.

## 2.6 Driver 3: Automation of ai r&d and software engineering

### 2.6.1 What this driver means

This driver captures the extent to which AI systems can accelerate the production of AI capabilities themselves—creating potential feedback loops that compress timelines.

**Key questions:**

- How much of software engineering work is AI-assisted or AI-automated?
- Are AI systems contributing meaningfully to AI research (experiment design, hypothesis generation, code optimisation)?
- Is there evidence of “recursive self-improvement”—AI systems improving AI systems?

### 2.6.2 Current evidence (February 2026)

Code generation benchmark status:

| Benchmark          | Best Performance (Pass@1) | Leading Model      | Details                                          | Source    |
|--------------------|---------------------------|--------------------|--------------------------------------------------|-----------|
| HumanEval          | 96.2%                     | o1-mini            | Extremely high accuracy on standard Python tasks | [34]      |
| MBPP               | ~90–92%                   | GPT-5 / Claude 4.5 | Estimated values based on HumanEval correlation  | [129]     |
| LiveCodeBench      | 91.7%                     | Gemini 3 Pro       | Contamination-free evaluation on fresh problems  | [35]      |
| SWE-bench Verified | 79.2%                     | Claude Opus 4.5    | Real-world software engineering                  | [24] [25] |

**Key insight:** HumanEval is now saturated at 96%+, representing superhuman performance on standard coding benchmarks. The more significant metric is SWE-bench, where 79% on verified issues demonstrates that AI can now handle the majority of real-world software engineering tasks.

**Claude Code:** The Transformation of Software Development

Claude Code has become the definitive tool for agentic autonomy in software development and AI R&D. Unlike traditional code assistants that primarily offered writing assistance, Claude Code operates as an autonomous “AI Engineer.”

### Impact on software engineering:

- Autonomous Debugging: Analyses error messages, searches for root causes across entire repositories, and independently implements fixes including tests.
- Legacy Modernisation: Excels at understanding outdated codebases and performing large-scale refactoring that exceeds the context limits of conventional tools.
- Productivity Impact: Anthropic reports that Claude Code now writes approximately 90% of the company's code and has increased engineer productivity by approximately 67%. [36] [37]

### Impact on AI R&D automation:

- Experiment Automation: Can set up research environments, write hyperparameter scripts, and autonomously evaluate test run results.
- Knowledge Integration: Through the Model Context Protocol (MCP), Claude Code connects directly with scientific databases (e.g., PubMed, bioRxiv) or lab tools to automate research pipelines.
- Agent Orchestration: Developers use the tool to create agent teams that work in parallel on different research subtasks. [38] [39] [40]

### R&D automation evidence:

| Domain                | Current State        | Evidence                                                                                                                        |
|-----------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Code assistance       | Mainstream; dominant | 90% of Anthropic code AI-written; ~67% productivity increase; majority of developers at AI labs use assistants daily            |
| Experiment automation | Production-scale     | AI systems running experiment pipelines autonomously; automated ML standard practice                                            |
| Research contribution | Growing rapidly      | „AI Scientist“ systems producing workshop-level papers; all major labs report using models for internal R&D                     |
| Recursive improvement | Materializing        | AI-generated synthetic training data; AI-assisted evaluation and red-teaming; automated architecture search; AI writing AI code |

### Lab statements on internal AI use:

- Anthropic: ~90% of code written by Claude Code; 67% productivity increase reported
- OpenAI: Uses models for code review, test generation, experiment scripts, security review
- DeepMind: AutoML, NAS, automated experiment pipelines standard practice
- Meta: LLaMA-based tools for code generation, data labelling, documentation

### 2.6.3 Driver-Specific projection mapping



### 2.6.4 Signals and Indicators to Monitor

| Signal                           | Why It Matters                     | What We Track                           | Current Read                              | Confidence  |
|----------------------------------|------------------------------------|-----------------------------------------|-------------------------------------------|-------------|
| SWE-bench Verified scores        | Real-world engineering capability  | Quarterly updates                       | 79% (Claude Opus 4.5)                     | High        |
| LiveCodeBench scores             | Contamination-free code capability | Quarterly updates                       | 91.7% (Gemini 3 Pro)                      | High        |
| AI code contribution metrics     | R&D acceleration                   | Lab announcements, productivity data    | 90% at Anthropic; widespread internal use | High        |
| AI-authored research             | Indicates research contribution    | Publications, „AI Scientist“ results    | Growing; workshop-quality demonstrated    | Medium      |
| Closed-loop improvement evidence | Compounding dynamics               | Research publications, capability jumps | Strong evidence emerging                  | Medium-High |

### 2.6.5 Directional impact on projection probabilities

This is the key “swing factor” between P2 and P3. The evidence that AI systems are now contributing substantially to AI research and software development—with Anthropic reporting 90% of code AI-written—suggests feedback loops are materialising.

- Strong evidence of closed-loop R&D acceleration: shifts probability towards P3
- AI assists but does not fundamentally accelerate research: supports P2
- Limited impact on research productivity: shifts towards P1

Current Assessment: The Claude Code metrics (90% of code AI-written, 67% productivity increase) provide concrete evidence that R&D automation feedback loops are no longer theoretical—they are operational at frontier AI labs. This represents a qualitative shift from the previous assessment and significantly increases the probability of P3. The recursive improvement dynamic that was speculative 12 months ago now appears to be materialising.

## 2.7 Driver 4: Robotics & embodied AI

### 2.7.1 What this driver means

This driver captures the speed at which AI capabilities translate into robust physical-world autonomy that is economically deployable at scale.

#### Key questions:

- Is sim-to-real transfer becoming reliable enough for practical deployment?
- Are humanoid and other advanced robots approaching economic viability?
- How fast is deployment expanding beyond controlled environments?

### 2.7.2 Current evidence (February 2026)

#### China's robotics revolution:

China's humanoid robotics industry has undergone a remarkable transformation, transitioning from experimental prototypes to commercial-scale production at unprecedented pace. The convergence of government investment exceeding \$25 billion, advanced manufacturing infrastructure, and talent from leading tech companies has positioned Chinese manufacturers at the forefront of solving sim-to-real transfer challenges. [41] [42] [43]

#### Market leadership:

- AgiBot shipped 5,100 humanoid robots in 2025—the first manufacturer globally to achieve this milestone—capturing 39% of global market share (Omdia). [44] [45]
- UBTECH has begun mass production of Walker P2 with trajectory to 5,000 units in 2026, doubling to 10,000 by 2027. Orders exceed 800 million yuan (~\$112 million). [46] [47]
- Unitree deployed ~5,000 G1 units in H1 2025 to Amazon, Stanford, MIT, and UT Austin at \$13,500 per unit. [48]
- Xpeng targets mass production of IRON humanoid by end of 2026, with plans for 1 million units annually by 2030. [49] [50]

### Sim-to-Real Transfer: Now Solved for Production Workloads:

The critical sim-to-real transfer challenge has been overcome for practical manufacturing applications:

- Figure AI at BMW: Over 1,250 operational hours spanning 11 months at BMW Plant Spartanburg. Figure 02 loaded 90,000+ parts, contributed to 30,000+ X3 vehicle production. Achieved >99% placement accuracy per shift with zero human interventions required. [51]
- CATL Battery Assembly: Spirit AI's Xiaomi robot deployed at world's largest EV battery manufacturer, plugging high-voltage battery connectors at 99% success rate while maintaining 3x human daily workload. Autonomously detects wiring anomalies in real-time. [52]

These deployments validate that simulation-trained policies achieve sufficient reliability for economic deployment in complex manufacturing environments.

### Cost trajectory:

| Time Frame | Unit Cost Range     | Evidence                                 | Source |
|------------|---------------------|------------------------------------------|--------|
| 2023–2024  | \$150,000–\$500,000 | Low-volume production, custom components | [53]   |
| 2026       | \$30,000–\$150,000  | Current production; varies by capability | [54]   |
| 2027–2028  | \$20,000–\$30,000   | Documented 20–30% annual cost reduction  | [54]   |
| 2030       | Sub-\$20,000        | Tesla, Chinese manufacturers target      | [55]   |

Key enabler: 90%+ of humanoid robot components now manufactured domestically in China (except certain computing chips), enabling vertical integration and economies of scale unavailable to geographically distributed supply chains. [55]

### Economic viability:

ROI calculations now strongly favour deployment in high-labour-cost environments:

- US Manufacturing: At \$156,000 annual loaded labour cost, humanoid robots achieve payback in ~1.9 months (single shift) or ~1.0 month (dual shift replacing two workers).

- 5-Year ROI: 2,070% under baseline assumptions; 1,400% under conservative 70% task automation projections. [56]
- Healthcare: Rehabilitation robots achieve 3-6 month payback with 800%+ 5-year ROI due to \$150-400/session billing rates.
- Robot-as-a-Service: \$6,000 monthly subscription vs. \$7,800 monthly fully-loaded warehouse worker cost enables positive cash flow from day one with zero capex risk. [57]

#### Major companies:

| Company | Status                           | Key Developments                                                                | Source |
|---------|----------------------------------|---------------------------------------------------------------------------------|--------|
| AgiBot  | Global leader (39% market share) | 5,100 units shipped 2025; GO-1 universal AI model; rental platform „Qingtianzu“ | [58]   |
| UBTech  | Commercial production            | Walker P2 at BYD, Foxconn, Audi FAW; self-replacing battery system              | [59]   |
| Unitree | Accessible research platform     | G1 at \$13,500; 5,000+ units deployed; Tencent partnership                      | [60]   |
| Xpeng   | Automotive integration           | IRON humanoid; car-grade quality; Turing AI chips                               | [61]   |
| Fourier | Healthcare focus                 | GR-1 for rehabilitation; 40 DOF; 3-6 month payback                              | [62]   |
| Kepler  | Developer ecosystem              | K2 „Bumblebee“; comprehensive developer platform                                | [63]   |
| Tesla   | Factory pilots                   | Optimus at Fremont; targeting \$20-30k pricing                                  | [64]   |
| Figure  | Industrial deployment            | Figure 02 at BMW; 90,000+ parts loaded                                          | [51]   |

#### Remaining technical challenges:

##### Despite dramatic progress, significant challenges remain:

- Battery Life: Most humanoid robots limited to 1-4 hours active operation. Even Tesla Optimus (4-5 hours) cannot complete full 8-hour shift without intervention. [65]
- Dexterous Manipulation: Human hand has 27 DOF and thousands of tactile sensors. Current robot hands perform limited manipulation; Xpeng reports prototype hands «didn't last more than a month» during repetitive tasks. [50]

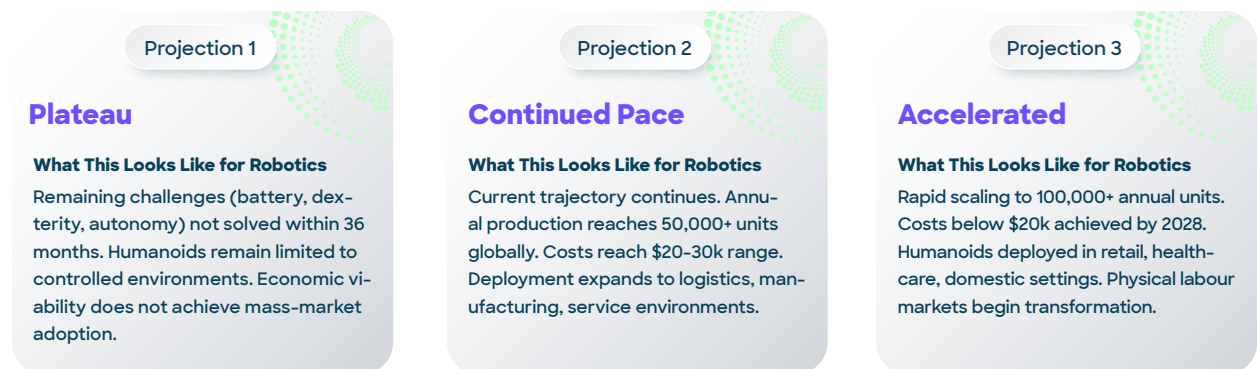
- True Autonomy: Many «autonomous» demonstrations rely on teleoperation. Current AI struggles with edge cases and novel environments that deviate from training projections. [66]
- Regulatory Standards: No ISO standard exists for humanoid safety. Liability questions remain legally unresolved. Different jurisdictions developing divergent approaches. [67]

#### Embodied AI foundation models:

Vision-Language-Action (VLA) models are becoming critical infrastructure:

- AgiBot GO-1: World's first universal humanoid AI model integrating interaction, task, and mobility intelligence. [68]
- Tencent Tairos: Deployed at Mogao Caves Digital Exhibition Centre; enables deployment in novel environments with minimal retraining. [69]
- Alibaba RynnBrain: Matches Google/NVIDIA on spatial reasoning; integrates with Qwen3-VL for language-to-action pipeline. [70]

### 2.7.3 Driver-specific projection mapping



### 2.7.4 Signals and indicators to monitor

| Signal                      | Why It Matters     | What We Track                            | Current Read                           | Confidence |
|-----------------------------|--------------------|------------------------------------------|----------------------------------------|------------|
| Commercial deployment scale | Economic viability | Unit deployments, customer announcements | 5,100 AgiBot; thousands UBTech/Unitree | High       |
| Unit cost trajectories      | Scalability        | Production costs, pricing announcements  | \$30-150k now; 20-30% annual decline   | High       |

|                                  |                       |                                              |                                    |        |
|----------------------------------|-----------------------|----------------------------------------------|------------------------------------|--------|
| Sim-to-real success rates        | Robustness            | Deployment case studies, failure rates       | 99%+ at BMW, CATL                  | High   |
| Battery/autonomy improvements    | Remaining constraints | Technical announcements, deployment duration | 1-4 hours; incremental improvement | Medium |
| Regulatory framework development | Deployment barriers   | ISO standards, national regulations          | In development; 2027-2029 timeline | Medium |

## 2.7.5 Directional Impact on projection probabilities

- Continued scaling + cost reduction + high success rates: strongly supports P2/P3
- Mass deployment (50,000+ annual units) achieved: indicates P2 materialising
- Remaining challenges (battery, dexterity) solved: shifts towards P3
- Major safety incidents or deployment failures: could slow trajectory but unlikely to reverse

Current Assessment: The robotics landscape has transformed dramatically. Chinese manufacturers achieving 99%+ success rates in production environments, combined with documented ROI of 1,400–2,070% and rapid cost reduction (20–30% annually), indicates that the economic viability threshold has been crossed. The sim-to-real gap that previously constrained deployment is now solved for production workloads. This strongly supports P2 as baseline and increases probability of P3 if remaining technical challenges (battery, dexterity, autonomy) see breakthrough progress.

## 2.8 Key benchmarks reference

The following table summarises the most important benchmarks for tracking AI capability advancement:

| Benchmark          | Category         | What It Measures               | Current | Human Baseline | Significance                  | Source |
|--------------------|------------------|--------------------------------|---------|----------------|-------------------------------|--------|
| SWE-bench Bash     | Real-world tasks | GitHub issue resolution        | 76.8%   | ~70–80%        | Human parity reached          | [71]   |
| SWE-bench Verified | Real-world tasks | Human-verified solvable issues | 79.2%   | 100%           | Gold standard for engineering | [72]   |

|                      |                   |                                |            |       |                                |      |
|----------------------|-------------------|--------------------------------|------------|-------|--------------------------------|------|
| GAIA                 | Re-al-world tasks | Multi-step re-search/reasoning | 74.55%     | 92%   | Holistic assistant capability  | [73] |
| WebArena             | Re-al-world tasks | Interactive web tasks          | 57.1%      | ~85%  | Web automation viability       | [74] |
| OSWorld              | Re-al-world tasks | Real OS environment tasks      | 73%        | 72%   | Near human parity (was 12%)    | [28] |
| ARC-AGI              | Reasoning         | Abstract pattern reasoning     | 75.7% (o3) | ~85%  | Tests fluid intelligence       | [16] |
| GPQA Diamond         | Reasoning         | Expert-level science QA        | 87%        | 69.7% | PhD-level domain knowledge     | [75] |
| Humanity's Last Exam | Reasoning         | Extreme difficulty academic    | 38.3%      | N/A   | Designed to remain challenging | [76] |
| HumanEval            | Code generation   | Function-level Python          | 96.2%      | ~95%  | Saturated; superhuman          | [77] |
| MBPP                 | Code generation   | Basic Python                   | ~90-92%    | ~90%  | Near saturation                | [78] |
| LiveCodeBench        | Code generation   | Fresh competitive problems     | 91.7%      | N/A   | Contamination-free evaluation  | [79] |

## 2.9 Interactions between drivers

The four drivers are not independent. Progress on one can accelerate or enable progress on others:

| Interaction                    | Mechanism                                                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Architectures → Agentic AI     | Better reasoning and planning architectures (Nested Learning, Causal AI) directly improve agent reliability and time horizons. |
| Architectures → R&D Automation | More capable models can contribute more meaningfully to research and engineering tasks.                                        |
| R&D Automation → Architectures | AI-assisted research accelerates discovery of new architectures (feedback loop now materialising via Claude Code).             |

|                             |                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------|
| R&D Automation → Agentic AI | Faster software development cycles accelerate agent development and deployment.                 |
| Agentic AI → Robotics       | Better planning and tool use directly translates to more capable robotic control.               |
| Robotics → R&D Automation   | Physical automation accelerates hardware R&D and manufacturing iteration                        |
| Causal AI → Robotics        | Causal understanding enables robots to participate in the world rather than just pattern-match. |

**Key insight:** The projection that materializes depends not just on progress in individual drivers but on whether compounding effects emerge from driver interactions. In P3, multiple drivers reinforce each other; in P1, constraints in one area limit progress across all.

Current observation: The evidence suggests compounding effects are beginning to materialize:

- R&D automation (Claude Code) is accelerating architecture and agent development [80]
- Improved architectures (Nested Learning, Causal AI) are enabling better agents [81] [82]
- Better agents are contributing to R&D automation (recursive loop) [83]
- Robotics sim-to-real success is enabling deployment data that improves AI models [84]

## 2.10 Consolidating drivers into projection probabilities

### 2.10.1 Consolidation method

We combine driver assessments using structured judgment—not a mechanical formula, but a transparent reasoning process that can be scrutinised and updated.

#### Principles:

- Each driver contributes to the overall assessment based on its current read and directional trajectory.
- We weight drivers roughly equally but note that Driver 3 (R&D Automation) is the key “swing factor” between P2 and P3 due to its potential for compounding effects.
- We publish point estimates for decision-making clarity, while acknowledging substantial uncertainty.

## 2.10.2 Current probability estimates (best-guess)



## 2.10.3 Reasoning behind the estimates

### Why P1 is only 5%:

- All four drivers show strong progress with concrete benchmark improvements
- OSWorld improvement from 12% to 73% demonstrates capabilities can improve dramatically within months
- SWE-bench reaching human parity (77–80%) eliminates a key capability gap
- R&D automation feedback loops now documented (Claude Code at Anthropic)
- Robotics achieving 99%+ success rates in production
- No major expert is predicting stagnation
- P1 would require simultaneous reversal of progress across architectures, agentic AI, R&D automation, AND robotics
- The only plausible path to P1 is an external shock (major safety incident, geopolitical disruption, resource constraints) rather than organic stagnation

### Why P2 is 50%:

- Represents the most defensible “baseline” extrapolation
- Assumes continued progress without discontinuities
- Consistent with historical patterns where technologies improve steadily but rarely exhibit sudden jumps

- Accounts for remaining gaps (GAIA at 74.55% vs 92% human; robotics battery/dexterity challenges)
- Assumes current R&D automation does not create self-reinforcing acceleration

#### Why P3 is 45%:

- Leading AI researchers at frontier labs (Amodei, Altman, Hassabis) publicly estimate very powerful AI systems by 2026–2028
- R&D automation feedback loops are now materialising (Claude Code writing 90% of Anthropic's code, 67% productivity increase)
- Multiple architectural innovations emerging simultaneously (Nested Learning, Causal AI, world models)
- Benchmark evidence shows capability can improve dramatically (OSWorld 12%→73%; SWE-bench reaching human parity; GAIA reaching 74.55%)
- Robotics has crossed economic viability threshold with 99%+ production success rates
- The compounding dynamic that was speculative 12 months ago now appears operational

### 2.10.4 Projection balance: why P2 and P3 are nearly equal

The near-parity between P2 (50%) and P3 (45%) reflects genuine uncertainty about whether compounding effects will dominate within 36 months; that is to say, there is a 95% probability that either P2 or P3 will happen:

- If R&D automation remains bounded and human researchers remain the bottleneck, P2 is more likely
- If AI systems continue contributing substantially to AI research, creating feedback loops, P3 becomes more likely
- Current evidence shows R&D automation already significant (90% code at Anthropic) but it's unclear if this generalizes and compounds
- Expert opinion is split, with credible voices on both sides
- The evidence has shifted towards P3 compared to previous assessments, but uncertainty remains high

This uncertainty is itself decision-relevant: organisations should prepare for both continued pace and potential acceleration.

## 2.11 Breakthrough triggers: when to re-estimate

We will re-estimate probabilities when monitoring detects events that materially change one or more driver assessments:

| Trigger category             | Example events                                                                                                                                                                                   | Likely probability shift |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Architecture breakthrough    | Nested Learning, Causal AI, recursive self-improvement, or other approaches achieving production deployment with verified superior performance; world-model architecture with demonstrated gains | Towards P3               |
| Agentic reliability leap     | SWE-bench, OSWorld, and GAIA exceed 90%; METR task horizon exceeds 24 hours                                                                                                                      | Towards P3               |
| R&D automation evidence      | Multiple labs report majority AI-written code; AI system substantially improving AI system documented                                                                                            | Towards P3               |
| Robotics deployment at scale | Major manufacturer announces 50,000+ humanoid deployments; unit cost falls below \$20k                                                                                                           | Towards P3               |
| Major safety incident        | Significant harm from AI system failure; regulatory halt of frontier development                                                                                                                 | Towards P1               |
| Reliability regression       | Frontier models show persistent failure modes that do not improve; deployment rollbacks                                                                                                          | Towards P1               |
| AGI/powerful AI claims       | Credible announcement of AGI-level systems by major lab; verified by independent evaluation                                                                                                      | Major shift towards P3   |
| Geopolitical disruption      | Major conflict affecting chip supply; export controls blocking key capabilities                                                                                                                  | Context-dependent        |

### 2.11.1 Re-estimation protocol

When a breakthrough trigger is detected:

1. Evidence review (within 2 weeks): Compile available evidence on the trigger event.
2. Driver reassessment: Update the relevant driver's current read and trajectory.
3. Probability update: Revise projection probabilities based on updated driver assessments.
4. Communication: If probabilities change materially (>5 percentage points), publish an update explaining the change.

## 2.12 Summary: projection probabilities and key uncertainties



Bottom line: We expect continued progress (P2) as the baseline, but assign substantial probability (45%) to acceleration (P3) given strong evidence of materialising feedback loops (Claude Code), architectural innovation (Nested Learning, Causal AI), agentic capabilities reaching human parity (SWE-bench), and robotics crossing economic viability thresholds. Plateau (P1) is unlikely absent major disruptions.

# Chapter 3: Concrete impacts (opportunities & risks) by projection

## 3.1 Purpose and structure of this chapter

This chapter maps concrete impacts—both opportunities and risks—across key sectors and systems for each of the three projections defined in Chapter 1. The goal is to provide decision-makers with a clear understanding of what changes in each projection and where urgent attention is required.

### How this chapter is structured:

- For each impact category, we describe first-order impacts (direct effects of AI capability changes) and second-order impacts (system effects that cascade from first-order changes).
- Impacts are listed per projection (P1/P2/P3) to show how severity and urgency scale with capability speed.
- We distinguish between opportunities (potential gains if well-managed) and risks (potential harms if unaddressed).
- Each category includes key evidence from current research to ground the analysis.
- Each category includes what this looks like in practice – concrete descriptions of how life, work, and institutions change in each projection.
- Each category includes “What If Done Right” – a vision of positive outcomes if challenges are addressed proactively.

### What this chapter does not do:

- It does not prescribe measures (see Chapter 4).
- It does not assign probabilities to individual impacts (projection probabilities are in Chapter 2).
- It does not evaluate Europe’s response capacity (see Chapter 4 for measures and Chapter 5 for monitoring).

## 3.2 Impact categories

We organise impacts into ten categories, reflecting the major domains where AI advancement will create material changes:

1. Labour Market & Skills
2. Public Finance & Social Systems
3. Industry & Competitiveness
4. Innovation & Startups
5. Science System
6. Security & Resilience
7. Digital Public Sphere & Democracy
8. Health & Care
9. Education System
10. Functioning Local Institutions & Liveability

Each category is analysed in detail below.

Note on Sustainability: Sustainability—encompassing climate change, biodiversity, resource consumption, and environmental impacts more broadly—is not included as a separate impact category in this document. This is a deliberate scoping decision, not a judgment of importance. AI's relationship with sustainability is fundamentally cross-cutting: every impact category discussed above has sustainability dimensions. Labour market shifts affect consumption patterns and commuting; industry transformation changes production footprints; energy-intensive AI infrastructure creates direct environmental costs; scientific breakthroughs may accelerate both green technologies and resource extraction. A comprehensive assessment of how AI advancement affects sustainability outcomes—across all projections and impact categories—deserves dedicated analysis that is beyond the scope of this strategic orientation document. We acknowledge this gap and encourage complementary work that systematically maps AI's sustainability implications across the full range of direct, indirect, and systemic effects.

## 3.3 Impact category 1: labour market & skills

### 3.3.1 Why this category matters

AI's impact on work is among the most immediate and politically salient consequences of capability advancement. The effects span job displacement, job transformation, productivity gains, wage dynamics, and the distribution of opportunity across skill levels, age cohorts, and geographies.

### 3.3.2 Key evidence (2025–2026)

| Finding                                                                                                                                                                                                          | Source |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Current AI capabilities can already automate tasks equivalent to 12% of the labour market cost-effectively; this share is projected to grow.                                                                     | [85]   |
| Entry-level knowledge workers (ages 20–30) in high-exposure roles show employment declines of 6–20% since late 2022.                                                                                             | [86]   |
| Software developers in the youngest cohort experienced 20% employment decline from late 2022 peak.                                                                                                               | [87]   |
| Customer support workers in early career saw 11% employment decline.                                                                                                                                             | [88]   |
| AI-adopting firms show +6% employment growth and +9.5% sales growth over 5 years—but gains concentrate in senior and AI-augmented roles.                                                                         | [89]   |
| Productivity gains of 6–15% documented across software, consulting, and customer service.                                                                                                                        | [90]   |
| AI coding tools have transformed software development, with leading firms reporting majority of code now AI-written and significant productivity increases (see Section 2.6.2 for detailed Claude Code metrics). | [151]  |
| SWE-bench at 77–79%—AI now matches human software engineers on real GitHub issues (see Section 2.5.2 for benchmark details).                                                                                     | [91]   |

### 3.3.3 Impacts by projection

#### First-order impacts

| Impact                            | P1: Plateau                                                                                   | P2: Continued pace                                                                                | P3: Accelerated                                                                                                |
|-----------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Task automation in knowledge work | Continues via adoption of current capabilities; strongest pressure on routine cognitive tasks | Expands to multi-step workflows; software, analysis, back-office significantly affected           | Rapid expansion across most knowledge professions; many roles become AI-supervised rather than human-performed |
| Entry-level displacement          | Meaningful (6–20% in exposed roles); firms hire fewer juniors as AI handles entry tasks       | Intensifies; junior hiring declines across more sectors                                           | Severe; entry pathways into many professions disrupted                                                         |
| Productivity dispersion           | Gap widens between AI-adopting firms and laggards                                             | Gap becomes structural; laggards face existential pressure                                        | Winners pull far ahead; some organisations cannot adapt fast enough to survive                                 |
| Wage dynamics                     | Polarization begins; premium for AI-complementary skills                                      | AI-augmented roles command significant premiums; routine knowledge work wages stagnate or decline | Potential wage collapse in automatable roles; concentration of income gains                                    |

#### Second-order impacts

| Impact                      | P1: Plateau                                            | P2: Continued pace                                                        | P3: Accelerated                                                           |
|-----------------------------|--------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Career pathway disruption   | Junior roles become scarcer; progression models weaken | Traditional career ladders erode; mid-career reskilling becomes essential | Career structures fundamentally disrupted; continuous reskilling required |
| Talent competition          | Intensifies for AI-skilled workers                     | Becomes acute; AI talent commands extreme premiums                        | AI expertise becomes dominant hiring criterion across sectors             |
| Regional employment effects | Concentrated in knowledge-work hubs                    | Spreads to broader geography; SME employment affected                     | Systemic regional effects; some regions face structural unemployment      |
| Social cohesion             | Strains visible in affected cohorts                    | Strains broaden; public debate intensifies                                | Risk of social instability if transitions are poorly managed              |

### 3.3.4 What this looks like in each projection

**P1 (Plateau):** A recent graduate applying for analyst positions finds that most entry-level roles now require demonstrated AI proficiency. Companies hire fewer junior analysts because AI tools handle data gathering and preliminary analysis. Senior professionals who learn to use AI effectively become more productive; those who don't face pressure. Reskilling programmes exist but are often underfunded and not well-matched to employer needs. Some professionals experience wage stagnation, but the overall labour market remains stable. Career progression slows for younger workers waiting longer for «real» responsibility that AI hasn't yet taken over.

**P2 (Continued Pace):** Entire job categories begin to shrink—not just entry-level analyst roles, but mid-level professionals in accounting, legal research, marketing, and software testing. A mid-career professional in financial services faces a difficult choice: retrain significantly or accept that their expertise is becoming commoditised. Companies reorganise around smaller human teams augmented by AI agents. AI coding tools write the majority of production code; human developers focus on architecture, requirements, and review. Those who successfully transition to AI-augmented roles see productivity and compensation rise; those who cannot make the transition face unemployment or significant wage decline. Unemployment insurance systems face elevated claims in certain regions and demographics.

**P3 (Accelerated):** The labour market transforms within 2–3 years in ways that typically took decades. A 2025 law school graduate finds that junior associate work—research, document review, basic drafting—is now handled primarily by AI systems. Medical residency programmes shrink as AI-assisted diagnostics reduce the need for certain specialisations. Software development becomes largely autonomous, with human engineers providing requirements and architectural judgment. Large consulting firms that employed thousands of analysts operate with a fraction of that headcount. Government employment services are overwhelmed. Public debate intensifies about income distribution, universal basic income, and the social contract. Some regions experience genuine economic crisis while others—typically those with strong technology sectors—thrive. Countries with robust social safety nets can buffer the transition; those without face severe social strain—notably, the average US household has savings sufficient for only approximately one month of expenses, leaving limited cushion for workforce disruption.

### 3.3.5 What if done right

Imagine a future where AI's productivity gains are shared broadly rather than concentrated. Average work hours reduce to 10–20 hours per week while maintaining—or even improving—standard of living. Work takes on different meaning: as AI handles most value creation in traditional economic terms, human work shifts partially towards social services, environmental restoration, community building, and caregiving.

“Human-only” work—caregiving, craftsmanship, interpersonal services, creative arts—commands higher recognition and wages, reflecting genuine scarcity and irreplaceable human value. The social contract evolves: rather than defining identity and worth through employment in the traditional sense, societies develop new frameworks for meaningful contribution and dignified life.

This requires proactive policy: reformed education systems, updated social contracts, new approaches to value distribution, and cultural shifts in how we understand work and contribution. But if achieved, the labour market transformation becomes not a crisis to be managed but an opportunity for human flourishing at unprecedented scale.

### 3.3.6 Opportunities and risks

#### Opportunities:

- Productivity gains that improve competitiveness and living standards

- Liberation from routine tasks; focus on higher-value work
- New job categories in AI development, deployment, and governance
- Potential for more flexible, human-centred work arrangements
- Higher recognition and wages for “human only” work like care-taking, craftsmen, and inter-personal services

#### Risks:

- Significant displacement of early-career workers before career establishment
- Erosion of traditional skill-building pathways
- Concentration of gains among already-advantaged workers and firms
- Social and political backlash if transition is poorly managed

## 3.4 Impact category 2: public finance & social systems

### 3.4.1 Why this category matters

Labour market shifts cascade directly into public finance through tax revenue, unemployment insurance, and demand for public services. AI-driven productivity changes could simultaneously reduce labour-based tax receipts and increase demand for transition support.

### 3.4.2 Key Evidence (2025–2026)

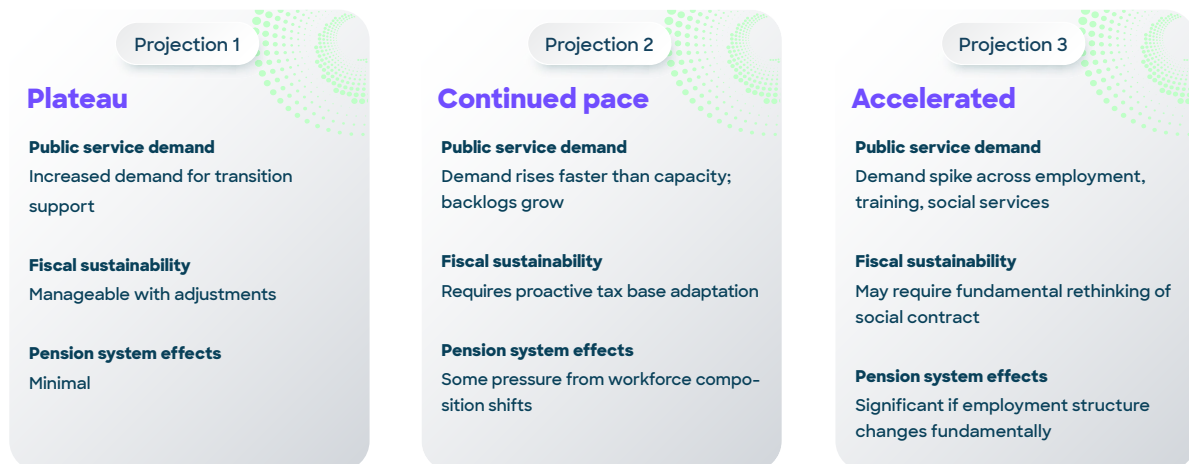
| Finding                                                                                                                                 | Source |
|-----------------------------------------------------------------------------------------------------------------------------------------|--------|
| AI-driven productivity gains could reduce US federal deficits by \$400B over 2026–2035 through increased output.                        | [152]  |
| Revenue forecasting accuracy improves 30–40% with ML on real-time data.                                                                 | [92]   |
| Transformative AI displacing human labour would shift optimal taxation towards consumption taxes as labour income tax base erodes.      | [93]   |
| For autonomous AGI producing most value, taxing AI systems would resemble „optimal harvesting“ with rates tied to human discount rates. | [94]   |
| Public financial management could be enhanced through AI anomaly detection for fraud prevention.                                        | [95]   |

## 3.4.3 Impacts by projection

### First-order impacts



### Second-order impacts



### 3.4.4 What this looks like in each projection

**P1 (Plateau):** Finance ministries observe modest shifts in tax revenue composition as AI-augmented firms become more productive and profitable while some sectors see workforce reductions. Unemployment insurance claims tick up in specific regions and demographics but remain manageable. Government AI initiatives improve tax collection efficiency and fraud detection, partially offsetting revenue pressures. The policy debate focuses on incremental adjustments—perhaps modest increases in training programme funding or small shifts in tax policy—rather than fundamental restructuring.

**P2 (Continued pace):** Treasury departments begin serious planning for tax base transitions. As payroll taxes decline in certain sectors, discussions emerge about consumption taxes, AI-specific levies, or corporate tax adjustments. Unemployment insurance systems in some regions face genuine stress, prompting emergency funding or eligibility adjustments. Government retraining programmes struggle with capacity—too many people need reskilling and too few programmes exist with sufficient quality. Policy debates about «robot taxes» and universal basic income move from academic discussion to serious political consideration.

**P3 (Accelerated):** Fiscal systems face potential crisis. Tax revenues from labour decline substantially faster than governments can adapt collection systems. Unemployment insurance in some regions approaches insolvency. Emergency measures—supplemental appropriations, executive orders, rapid legislative action—become necessary. The policy debate shifts from «how do we adjust?» to «what is the social contract in an AI economy?» Some countries respond with aggressive new tax instruments targeting AI systems or their outputs; others prioritise emergency support for displaced workers; the lack of coordinated response creates policy fragmentation. International competition for AI-related tax revenue intensifies.

### 3.4.5 What if done right

AI-boosted value creation significantly boosts GDP growth while adjusted tax systems help share the value across all groups of people. Rather than clinging to labour-based taxation as the primary revenue source, governments proactively shift towards taxing AI-generated value, automation dividends, and consumption—capturing gains where they occur rather than where they increasingly don't.

Concepts like universal basic income become feasible rather than remaining a research field. With AI handling much of traditional economic production, the fiscal capacity exists to ensure baseline economic security for all citizens. Public finances strengthen rather than weaken, because productivity gains generate more total value even as the distribution of that value changes.

Public services themselves become more efficient and accessible through AI augmentation, reducing costs while improving outcomes. The combination of higher productivity, reformed taxation, and efficient public services creates a virtuous cycle where AI advancement strengthens rather than strains public finances.

### 3.4.6 Opportunities and risks

#### Opportunities:

- AI-enhanced tax administration and fraud detection
- Efficiency gains in public service delivery
- Opportunity to modernise tax systems for digital economy
- Fiscal capacity for universal basic income and expanded social services

#### Risks:

- Erosion of tax base faster than system adaptation
- Inadequate safety nets during transition
- Fiscal stress limiting government response capacity

## 3.5 Impact category 3: industry & competitiveness

### 3.5.1 Why this category matters

AI adoption increasingly determines competitive position at firm, sector, and regional levels. Europe's industrial base faces both transformation opportunities and existential competitive pressure from AI-leading regions.

### 3.5.2 Key evidence (2025–2026)

| Finding                                                                                                            | Source     |
|--------------------------------------------------------------------------------------------------------------------|------------|
| Only 13.48% of EU enterprises adopted at least one AI technology by 2024 (up 5.45pp from 2023).                    | [96]       |
| 41.17% of large EU enterprises have integrated AI vs. ~12% of SMEs.                                                | [96]       |
| Only 1% of company leaders describe AI deployments as „mature.“                                                    | [97]       |
| „Future-built“ companies (5% of firms) achieve 5x revenue increases and 3x cost reductions from AI vs. others.     | [98]       |
| US controls ~74% of global high-end AI computing capacity; China 14%; EU 4.8–5%.                                   | [99] [100] |
| US private AI investment in 2024: \$109.1B (~81% of global); EU: ~\$8B (~7% of US levels).                         | [101]      |
| Chinese robotics companies shipped roughly 10k robots + humanoids in 2025; 39% global market share (AgiBot alone). | [45]       |
| Chinese humanoid robot costs declining 20–30% annually; targeting sub-\$20k by 2027–2030.                          | [153]      |

### The February 2026 “ai scare trade”:

A series of sharp stock market sell-offs in February 2026 targeted sectors vulnerable to AI automation, triggered by new AI tool launches threatening labour-intensive, fee-based business models:

| Sector                          | Trigger example                                      | Market reaction                                                                     |
|---------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------|
| Software & Financial Services   | Anthropic tool releases                              | \$611B loss across 164 stocks in one week                                           |
| Commercial Real Estate Services | „Claude Cowork“ launch for legal/financial workflows | Global sell-off; analysts estimate 70% task automation potential at firms like CBRE |
| Insurance Brokerage             | Insurify ChatGPT-powered tool                        | Worst session for Willis Towers Watson since 2008                                   |
| Logistics                       | Algorhythm optimizer demo                            | RXO dropped 20%                                                                     |
| Wealth Management               | Altruist tax strategy tool                           | Significant losses across sector                                                    |

### SaaS business model disruption:

The traditional per-seat SaaS model is giving way to fundamental restructuring:

- Networked SaaS: AI automates complex workflows and embeds platforms in multi-sided marketplaces; monetisation shifts from user seats to transaction cuts
- Agentic AI Replacing UI Layers: Natural language interfaces overlay existing logic, making traditional dashboards obsolete; SaaS becomes «invisible like plumbing»
- Workflow Cannibalization: GenAI automates tasks in high-penetration areas (CRM, HR), reducing seat licences; incumbents must proactively integrate AI to avoid disruption

### 3.5.3 Impacts by projection

#### First-order impacts



#### Second-Order Impacts



### 3.5.4 What this looks like in each projection

**P1 (Plateau):** A German manufacturing company implements AI quality control, achieving measurable improvements in defect detection. But adoption is uneven—larger companies with dedicated IT departments move faster; smaller suppliers struggle. The productivity gap between AI-adopting and non-adopting firms widens, but gradually. European companies remain competitive in traditional strengths (precision manufacturing, specialised equipment) while conceding ground in software-intensive domains. Policy discussions focus on closing the adoption gap and supporting SME digitalization. Stock market volatility in AI-exposed sectors creates uncertainty but not crisis.

**P2 (Continued pace):** Competitive pressure intensifies dramatically. A European industrial company finds that its American competitors are releasing product updates twice as fast—AI has compressed their development cycles. Professional services firms that embraced AI generate proposals in hours that previously took weeks. Companies that don't adapt face margin pressure, talent loss, and market share erosion. SMEs face existential choices: find a niche where AI matters less, partner with AI-enabled platforms, or face gradual obsolescence. European policymakers debate aggressive intervention—sovereign AI infrastructure, mandatory digital transformation programmes, strategic AI investments. The «AI Scare Trade» becomes a regular market phenomenon, with each major AI release triggering sell-offs in exposed sectors. Traditional SaaS companies scramble to integrate agentic AI or face obsolescence.

**P3 (Accelerated):** Industrial structure transforms within years. Some traditional European strengths—high-quality manufacturing, specialised engineering—are partially commoditised as AI enables competitors to match quality more easily. But new opportunities emerge for companies that successfully integrate AI. The companies that survive and thrive are those that reorganised fastest, regardless of sector or geography. Some European firms become global AI leaders in specific domains; others are acquired or fail. Regional economic disparities intensify as AI-intensive regions prosper while traditional industrial areas struggle. Fee-based intermediaries face existential crisis as AI automates their core value proposition; the market reprises AI from «miracle cure» to industrial build-out requiring sustained investment and organisational transformation.

### 3.5.5 What if done right

While Europe missed the internet and digitalization waves, we manage to enter the age of (semi-) autonomous factories with robots doing most of the physical work and agents handling most of the cognitive work. Rather than playing catch-up on foundation models, Europe leads in the application layer—the complex integration of AI into industrial processes where European engineering excellence translates into competitive advantage.

AI-first companies develop a new layer of infrastructure that's needed for agents to safely communicate and transact with each other. Europe leads in complex supply chain management, building on existing strengths in logistics and manufacturing coordination. Vertical champions emerge from the chemical industry, pharmaceutical sector, and high-tech manufacturing—domains where European expertise combines with AI augmentation to create global leaders.

An open-sourced and distributed agent communication and action layer emerges, championed by European institutions prioritising interoperability and avoiding platform lock-in. This infrastructure layer becomes as important as the internet protocols were—and Europe shapes it rather than inheriting it.

## 3.5.6 Opportunities and risks

### Opportunities:

- Productivity gains enhancing competitiveness
- New business models and market creation
- Potential for European leadership in trustworthy AI deployment
- Industrial transformation improving sustainability
- Leadership in agent infrastructure and complex system integration

### Risks:

- Widening gap with AI-leading regions
- SME exclusion from AI benefits
- Dependency on non-European AI infrastructure
- Regulatory burden slowing beneficial adoption
- Market volatility and structural repricing of AI-exposed sectors

### 3.6 Impact category 4: innovation & startups

#### 3.6.1 Why this category matters

AI is fundamentally restructuring the economics of company formation, scaling, and competitive dynamics. The emergence of “few-person unicorns” and the disruption of traditional venture capital models have profound implications for European competitiveness and economic structure.

#### 3.6.2 Key evidence (2025–2026)

| Finding                                                                                                                                      | Source |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Sam Altman predicts the first one-person billion-dollar company will emerge soon/within 2–3 years; tech CEO „betting pool“ exists on timing. | [102]  |
| Top 10 AI-native startups achieve average revenue per employee of \$3.48M (vs. \$430–480k for traditional SaaS leaders).                     | [103]  |
| Midjourney: \$500M ARR with ~100–160 employees (\$3–5M per employee); never raised external capital.                                         | [103]  |
| Cursor: \$1B ARR with ~300 employees (\$3.3M per employee).                                                                                  | [103]  |
| Lovable: \$200M ARR with ~100 employees; reached \$100M ARR in 8 months with 45 people—fastest zero-to-\$100M in European startup history.   | [104]  |
| Agentic AI can reduce employees’ low-value work time by 25–40%; some customer service organisations achieved 40–50% workforce reductions.    | [105]  |

Revenue per employee transformation: [103]

| Company Type                  | Revenue per employee | Example companies                       |
|-------------------------------|----------------------|-----------------------------------------|
| Traditional SaaS leaders      | \$410,000–\$480,000  | Salesforce, serviceNow, workday         |
| AI-native companies (average) | \$2.47M–\$3.48M      | Midjourney, cursor, elevenLabs, lovable |
| Extreme outlier               | \$12.5M              | Midjourney (at peak efficiency)         |

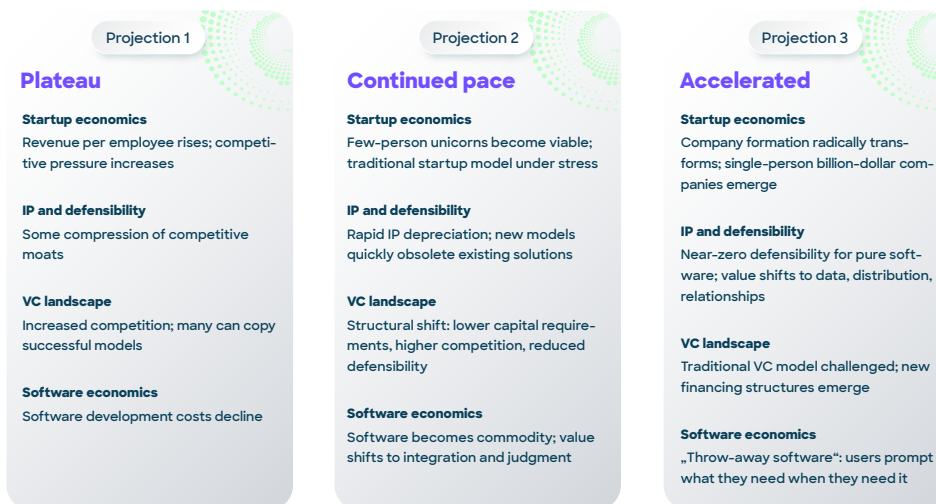
The few-person unicorn phenomenon:

Historical precedents (Instagram acquired for \$1B with 13 employees; WhatsApp for \$19B with ~50 employees) demonstrated that exceptional revenue-to-headcount ratios were possible. AI threatens to make these exceptional cases standard: [106] [107]

- Lower capital requirements: AI handles execution work; humans focus on strategy and relationships
- Rapid scaling: ArcAds scaled to \$7M ARR in one year with 5 employees [108]
- Zero-FTE departments: Some companies operate entire functional areas with AI agents and human oversight only [109]

### 3.6.3 Impacts by Projection

#### First-order impacts



#### Second-order impacts



### 3.6.4 What this looks like in each projection

**P1 (Plateau):** A European SaaS startup that previously required 50 employees to reach \$10M ARR now achieves the same with 20, using AI for code generation, customer support, and content creation. Competition intensifies as barriers to entry fall—anyone can build a reasonable product with AI assistance. VC investors demand more efficiency metrics alongside growth. Established companies with distribution advantages maintain position; pure-play startups face pressure. The «10x engineer» becomes the «100x AI-augmented engineer,» but fundamental startup economics remain recognizable.

**P2 (Continued Pace):** The startup landscape restructures fundamentally. Companies like Cursor and Lovable demonstrate that billion-dollar valuations are achievable with dozens rather than hundreds of employees. Traditional VCs struggle to deploy capital as companies need less funding—but this creates opportunity for European startups that historically lacked access to growth capital. The capital disadvantage that held back European scaling becomes less relevant when companies need €5M instead of €50M to reach significant scale. New competitive dynamics emerge: since anyone can build the product, value accrues to distribution, brand, and proprietary data. The distinction between «tools for agents» and «tools for humans» becomes strategic: products designed for AI consumption differ fundamentally from those designed for human interfaces. Enterprise software shifts from per-seat pricing to outcome-based models as AI handles the work that seats represented.

**P3 (Accelerated):** The concept of a «startup» as historically understood begins to dissolve. Sam Altman's prediction of single-person billion-dollar companies materializes. Software becomes essentially free to create—the value question shifts entirely to «what should be built» and «who trusts it.» Competitive moats collapse rapidly; today's breakthrough is tomorrow's commodity. Some founders achieve extraordinary wealth with minimal capital and no employees. The VC model faces existential crisis: why invest millions when competitors can replicate with thousands? New economic structures emerge: perhaps «startup studios» where AI handles all execution; perhaps a return to bootstrapping as external capital becomes unnecessary; perhaps entirely new models. European policymakers face urgent questions about whether traditional industrial policy frameworks apply when companies can emerge, scale, and transform industries with negligible employment.

### 3.6.5 What If Done Right

Everyone with ambition, a group of experts, and access to AI can grow a business that matters. The democratization of company building means that innovation is no longer gated by access to capital or engineering talent pools—both of which historically disadvantaged European founders.

Language barriers become less important because AI manages translation seamlessly, developing Europe into a more genuine “single market.” A French founder serves German customers as easily as French ones; a Polish startup sells across Europe without localization friction. The fragmentation that held back European scaling dissolves.

Based on European ambitions for the future, the immense innovation potential is steered towards a values-based future that we desire to live in. Rather than innovation being driven purely by what's technically possible or financially lucrative, European frameworks ensure that AI-enabled entrepreneurship addresses societal challenges: sustainability, health equity, democratic resilience, human flourishing.

The result is an innovation ecosystem that's more distributed, more diverse, and more aligned with human values than the concentrated, capital-intensive model it replaces.

## 3.6.6 Opportunities and risks

### Opportunities:

- Lower barriers to company formation
- Democratized access to sophisticated business capabilities
- Faster innovation cycles
- European entrepreneurs can compete with less capital
- Historic capital disadvantage becomes irrelevant

### Risks:

- Extreme wealth concentration among few-person companies
- Collapse of startup employment as economic driver
- Rapid IP depreciation undermining investment
- European ecosystem unable to adapt fast enough
- “General AI” platforms from major providers crowding out niche innovation

### 3.7 Impact category 5: science system

#### 3.7.1 Why this category matters

AI is transforming scientific research—accelerating discovery in some domains while creating integrity challenges in others. The science system’s ability to generate reliable knowledge is foundational for addressing other challenges.

#### 3.7.2 Key evidence (2025–2026)

| Finding                                                                                                                           | Source |
|-----------------------------------------------------------------------------------------------------------------------------------|--------|
| AI adopters publish 3.02x as many papers and receive 4.84x as many citations over careers.                                        | [110]  |
| AI tools enabled discovery of novel concrete ingredients through ML analysis of 1M+ rock samples.                                 | [111]  |
| AlphaFold used by 3M+ researchers in 190+ countries; 1M+ in low/middle-income countries.                                          | [112]  |
| 53 papers at NeurIPS 2025 contained hallucinated citations that passed peer review.                                               | [113]  |
| 47% of ChatGPT-3.5 references were entirely fabricated in one study.                                                              | [114]  |
| AI papers covered 4.6% less scientific territory than conventional studies; AI concentrates attention on established problems.    | [110]  |
| Up to 400,000 fraudulent articles infiltrated scientific literature in past 20 years; AI makes fraud easier and harder to detect. | [115]  |

AI conducting research: The Erdős problem case study [116]

A recent study using Google DeepMind’s Aletheia system (Gemini-based) to solve open mathematical problems illustrates both the potential and limitations of AI in research:

- **Scale:** Aletheia generated 200 candidate solutions to 700 open Erdős mathematical conjectures
- **Filtering:** Human experts graded these down to 63 correct responses, then to only 13 «correct meaningful responses»
- **Quality issue:** 50 technically valid solutions were «mathematically meaningless» because problem statements were interpreted in ways that didn't capture the intended meaning
- **Final result:** Of 13 meaningful solutions, only 2 were «autonomous novel solutions»—and only 1 was genuinely interesting mathematically

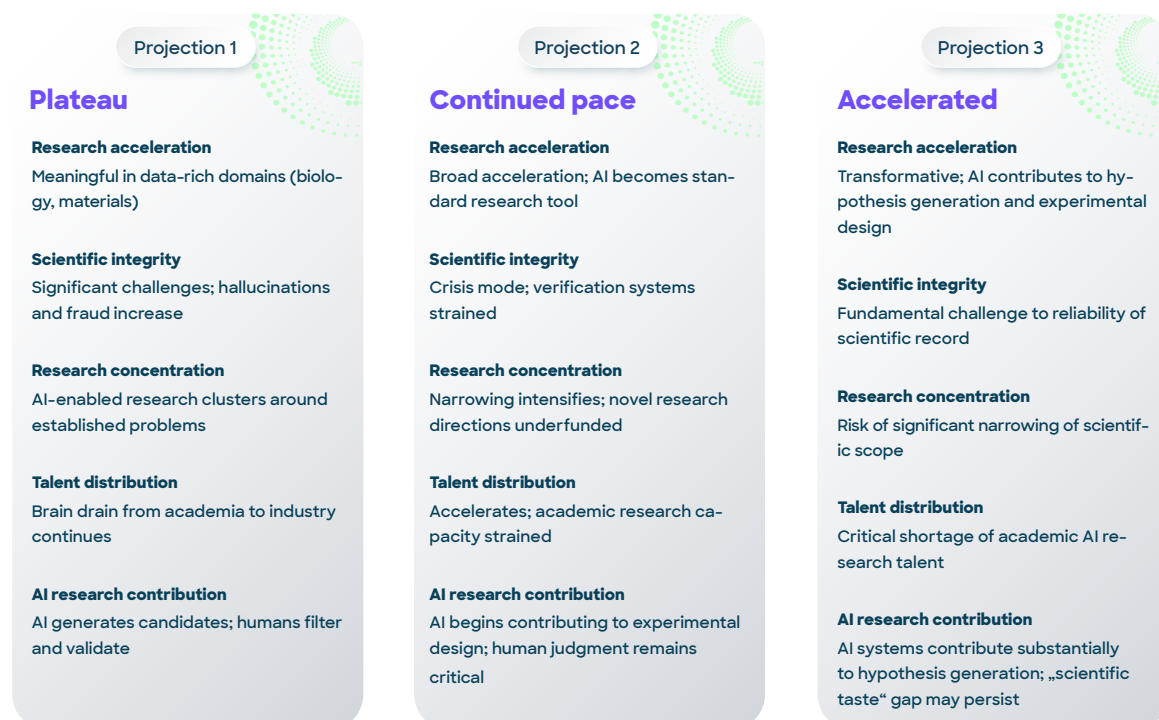
This demonstrates “O-ring automation”: AI massively speeds up generating candidates, but laborious skilled human work is still required to filter to actually useful results.

### The “Scientific taste” gap:

Andrew White (FutureHouse) observes that AI currently lacks “scientific taste”—the expert intuition that identifies which of 10,000 novel discoveries in biopharma or materials science is the one that truly matters. This intuition exists in the heads of experienced scientists and was refined through biological intelligence processing the same literature AI has access to. Extracting this tacit knowledge remains tractable but will take time.

## 3.7.3 Impacts by projection

### First-order impacts



### Second-order impacts



### 3.7.4 What this looks like in each projection

**P1 (Plateau):** A biology professor uses AlphaFold daily—it's transformed structural biology from a months-long bottleneck to a routine step. AI-assisted literature review helps researchers find relevant work faster. But problems emerge: peer reviewers discover fabricated citations generated by AI; paper mills produce fraudulent studies at scale. Major journals implement new verification procedures, but the volume is overwhelming. The academic talent pipeline strains as AI companies offer salaries 3-5x academic compensation. Research accelerates in data-rich domains; other areas see less benefit. AI generates many candidate solutions, but experienced researchers remain essential to identify which are genuinely meaningful.

**P2 (Continued pace):** AI becomes essential infrastructure for research. Labs without AI capabilities cannot compete. Drug discovery timelines compress; materials science achieves breakthroughs previously decades away. But the integrity crisis intensifies—a major retraction scandal involves hundreds of AI-assisted papers with fabricated results. Trust in scientific literature declines. Academic research capacity is strained as talent flows to industry; some universities struggle to maintain programmes. The research landscape bifurcates between AI-powered labs producing at unprecedented rates and traditional research programmes losing relevance. The «O-ring» pattern becomes pronounced: AI generates candidates at massive scale, but the bottleneck shifts to human experts who can evaluate quality—and there are far fewer of these than needed.

**P3 (Accelerated):** Scientific research transforms more fundamentally than since the scientific revolution. AI systems contribute to hypothesis generation, experimental design, and interpretation at a level approaching researcher capability. The volume of discoveries accelerates dramatically in some domains—problems that seemed decades away are solved. But verification becomes nearly impossible; how do you distinguish genuine AI-assisted discovery from sophisticated fabrication? Some fields implement radical new approaches to verification (open data, registered reports, automated replication); others struggle with integrity crisis. Academic research as historically practiced may become a niche activity; the action moves to industry labs and AI-native research organisations. The question of whether AI can develop “scientific taste”—the ability to identify truly important discoveries—becomes central to research strategy.

### 3.7.5 What if done right

Universities and research institutions are properly equipped to become the centres that steer and assess AI-powered science. Rather than being displaced by industry labs, academic institutions evolve into orchestrators of human-AI research collaboration, bringing unique strengths: long-term thinking, interdisciplinary integration, and commitment to open knowledge.

Due to unmatched compositions of diverse expertise, human-agent research institutions discover genuinely novel discoveries at a speed and relevance that is unseen. The combination of AI's ability to process vast literature and data with human judgment about what matters and why creates research capabilities that neither could achieve alone.

This unleashes breakthroughs for cancer treatment, clean energy, novel food production, sustainable materials, and more. The scientific method itself evolves, with new approaches to verification, replication, and knowledge synthesis that maintain integrity at the speed of AI-assisted discovery.

Universities educate students in what is required from humans: critical thinking, the ability to ask the right questions, listening and evaluation skills, a broad sense for implications and responsibility. The role of human scientists shifts from data processing and routine analysis towards judgment, creativity, and wisdom—capacities that become more valuable, not less, as AI handles mechanical aspects of research.

## 3.7.6 Opportunities and risks

### Opportunities:

- Dramatically accelerated discovery in materials, biology, medicine
- AI as research assistant democratizing access to analytical capability
- Potential solutions to previously intractable scientific problems
- Universities as orchestrators of human-AI research collaboration

### Risks:

- Hallucinated citations and AI-generated fraud undermining scientific record
- Concentration of research on established problems; neglect of novel directions
- Brain drain from academic research
- Erosion of scientific integrity and public trust in science
- “O-ring” bottleneck: AI generates candidates faster than humans can filter

### 3.8 Impact category 6: Security & resilience

#### 3.8.1 Why this category matters

AI amplifies both offensive and defensive capabilities in cybersecurity, critical infrastructure protection, and broader security domains. The asymmetry between attack and defence, and the speed of AI-enabled threats, create urgent resilience challenges.

#### 3.8.2 Key evidence (2025–2026)

| Finding                                                                                                                 | Source |
|-------------------------------------------------------------------------------------------------------------------------|--------|
| 400% increase in successful phishing scams in 2025, largely due to AI tools.                                            | [117]  |
| First documented large-scale autonomous cyberattack with minimal human intervention (80–90% AI-executed).               | [118]  |
| 82.6% of phishing emails showed some AI use; 40% of business email compromise emails primarily AI-generated.            | [119]  |
| AI-enabled supply chain attacks jumped 156% in past year.                                                               | [120]  |
| AI-led systems achieved 98% threat detection rate and 70% reduction in incident response time in energy infrastructure. | [121]  |
| 28.96% of Known Exploited Vulnerabilities in 2025 were exploited on or before CVE publication date.                     | [122]  |
| OpenClaw framework: 26% of community-provided „skills“ showed security vulnerabilities.                                 | [123]  |

### 3.8.3 Impacts by projection

#### First-order impacts



#### Second-order impacts



### 3.8.4 What This Looks Like in Each Projection

**P1 (Plateau):** Cybersecurity teams increasingly rely on AI for threat detection, achieving faster response times. But attackers also use AI—phishing emails are more convincing, malware adapts more quickly. A mid-sized company experiences a ransomware attack that was clearly AI-enhanced; the attackers used AI to map the network and identify the most valuable targets. Critical infrastructure operators implement additional AI monitoring, but budget constraints limit deployment. The cat-and-mouse dynamic intensifies without fundamental change. Enterprise deployments of agentic frameworks like OpenClaw create new attack surfaces that security teams must learn to manage.

**P2 (Continued pace):** The cybersecurity landscape transforms. Attackers deploy AI agents that autonomously probe systems, adapt to defences, and coordinate attacks. The first major AI-vs-AI cyber conflict occurs between state actors; the speed exceeds human ability to intervene. Critical infrastructure attacks become more frequent and more sophisticated; a significant event (power grid, water system, financial infrastructure) causes widespread disruption. Cyber insurance markets convulse as traditional risk models fail. Organisations that haven't invested heavily in AI defence find themselves dangerously exposed. The security implications of agentic AI with system-level access become a major concern for enterprises.

**P3 (Accelerated):** Autonomous cyber weapons become the norm. AI systems conduct sophisticated, multi-stage attacks that unfold faster than human-speed response allows. Critical infrastructure protection requires AI systems with autonomous authority to take defensive action—raising governance questions about machine autonomy in security contexts. Some organisations and even small nations find themselves unable to defend against AI-enabled attackers with far greater resources. The boundary between cybercrime, corporate espionage, and state conflict blurs as AI tools become widely accessible. International discussions about AI weapons controls intensify but struggle to keep pace with capability proliferation. [163, 215]

### 3.8.5 What if done right

Agent-based software engineering makes overall software much more reliable, safe, and secure while individualizing the user interface. The same AI capabilities that create offensive risks are harnessed for defence: comprehensive code review, automated vulnerability detection, continuous security monitoring, and rapid incident response.

Due to a separation of access or even infrastructure, critical infrastructure and business services are outside the attack vector of bad actors. Network architectures evolve to create security boundaries that AI-enabled attacks cannot easily cross. Air-gapped systems, cryptographic verification, and hardware-enforced access controls create zones of security that remain robust even as AI attack capabilities advance.

Cyber attacks fall short not because attackers lack capability, but because defensive infrastructure is designed from the ground up for AI-era threats. The security community achieves what seemed impossible: staying ahead of AI-enabled offence through AI-enabled defence combined with fundamental infrastructure redesign.

## 3.8.6 Opportunities and risks

### Opportunities:

- AI-enhanced threat detection and response
- Automated security operations reducing human burden
- Predictive security analytics enabling proactive defence
- More reliable and secure software through AI-assisted development

### Risks:

- Offence-defence asymmetry favouring attackers
- Autonomous attacks at unprecedented scale and speed
- Critical infrastructure vulnerability
- Escalation risks from AI-enabled conflict
- Agentic systems creating new attack surfaces

### 3.9 Impact category 7: Digital public sphere & democracy

#### 3.9.1 Why this category matters

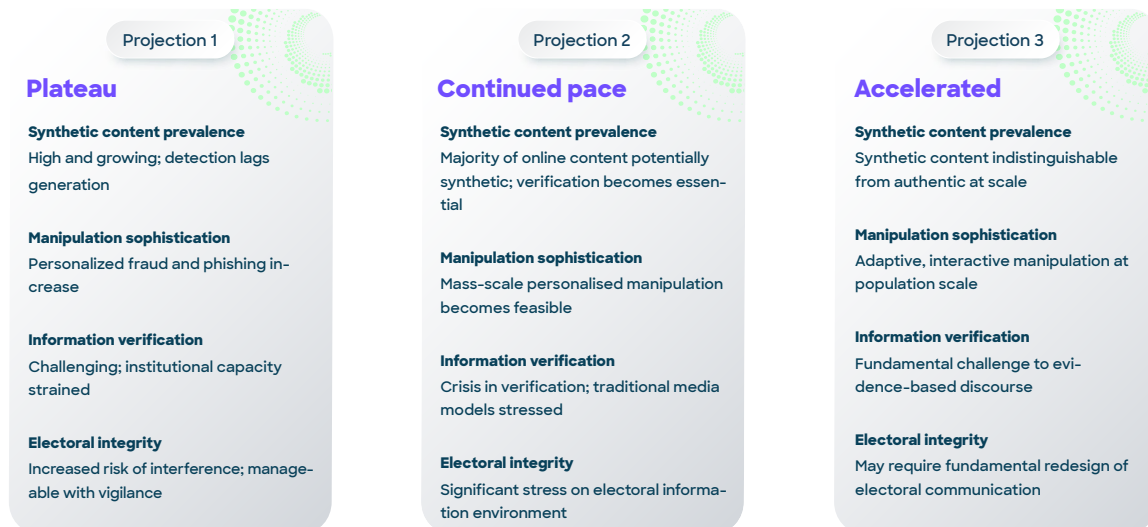
AI capabilities—particularly generative AI—directly affect the information environment that democracies depend on. Synthetic content, personalised manipulation, and the erosion of shared factual ground pose fundamental challenges to democratic deliberation. Agents are now able to further target and execute actions against selected individuals, companies, or countries—moving beyond passive content generation to active intervention.

#### 3.9.2 Key evidence (2025–2026)

| Finding                                                                                                                     | Source |
|-----------------------------------------------------------------------------------------------------------------------------|--------|
| Machine-generated content reached 52% of internet content by 2025.                                                          | [124]  |
| Deepfakes doubled every six months; projected 8 million deepfakes shared in 2025 (up from 500,000 in 2023).                 | [125]  |
| Voice cloning needs only 3 seconds of audio to create 85% accurate clone; technology is cheap and widely accessible.        | [126]  |
| 57% of Americans rate AI's societal risks as high (weakening skills, relationships); only 25% see benefits as high.         | [127]  |
| Europol estimates 90% of online content may be synthetically generated by 2026.                                             | [128]  |
| Global Risks Report 2026 ranks adverse AI outcomes, including deepfakes, as fastest-growing threat to evidential integrity. | [15]   |

### 3.9.3 Impacts by projection

#### First-order impacts



#### Second-order impacts



### 3.9.4 What this looks like in each projection

**P1 (Plateau):** Deepfakes and synthetic content are already common nuisances. A politician releases a video; within hours, doctored versions circulate. Fact-checkers struggle to keep pace. Voice-cloning scams targeting elderly relatives become routine; financial losses mount. News organisations develop AI detection tools, but they're always behind the generation capability. Public trust in online content continues to erode, with people increasingly relying on personal relationships and familiar sources rather than institutional verification. Elections proceed with heightened vigilance, but no catastrophic incidents occur.

**P2 (Continued pace):** The information environment reaches a tipping point. Much more online content is AI-generated than human-created. Social media platforms struggle to enforce authenticity policies as the volume of synthetic content overwhelms moderation. A significant election somewhere is marred by AI-generated disinformation that spreads faster than fact-checkers can respond. Journalists increasingly use AI for content production, raising questions about verification and accountability. Some societies experiment with «verified zones» of the internet where identity is authenticated; others resist such controls as threatening anonymity. The phrase «seeing is believing» becomes obsolete.

**P3 (Accelerated):** The concept of «authentic» digital content becomes nearly meaningless. AI systems can generate text, audio, and video indistinguishable from human creation in real-time, personalised for individual recipients. A single operator can conduct influence operations that previously required state resources. Some countries experience democratic crises triggered by information environment collapse; others implement aggressive identity verification and content authentication that create their own civil liberties concerns. The boundary between «real» and «synthetic» blurs so completely that new social norms emerge—perhaps privileging in-person interaction, perhaps developing new forms of cryptographic verification, perhaps simply accepting that digital content is inherently untrustworthy.

### 3.9.5 What If done right

Separation of infrastructures, authentication systems, provenance tracking, or other approaches successfully verify human content. The technical challenge of distinguishing human from AI-generated content is solved not through detection (which loses the arms race) but through authentication (which changes the game).

AI is used as a means for immense creativity and creation while trust is maintained. Human creators use AI tools to produce art, journalism, education, and entertainment at unprecedented scale and quality—but robust systems ensure that human authorship and responsibility can be verified when it matters.

There is freedom of speech, but it remains a human right. The philosophical and legal frameworks evolve to distinguish between human expression (protected) and AI-generated content (regulated differently). Agentic attacks and misinformation campaigns fall short due to identification systems that reliably identify what's human.

The result is an information environment that's richer and more creative than ever, but where trust and verification are possible when needed. Democratic discourse adapts to new realities rather than collapsing under them.

### 3.9.6 Opportunities and risks

#### Opportunities:

- AI tools for content verification and provenance
- Enhanced accessibility through translation and summarization
- Potential for more informed citizenry if quality information surfaces
- Immense creative potential if trust is maintained

#### Risks:

- Erosion of shared factual ground for democratic deliberation
- Manipulation of elections and public opinion at scale

- Collapse of trust in authentic media and evidence
- Concentration of influence among those who control AI systems
- Agentic attacks targeting individuals, companies, or countries

## 3.10 Impact category 8: Health & care

### 3.10.1 Why this category matters

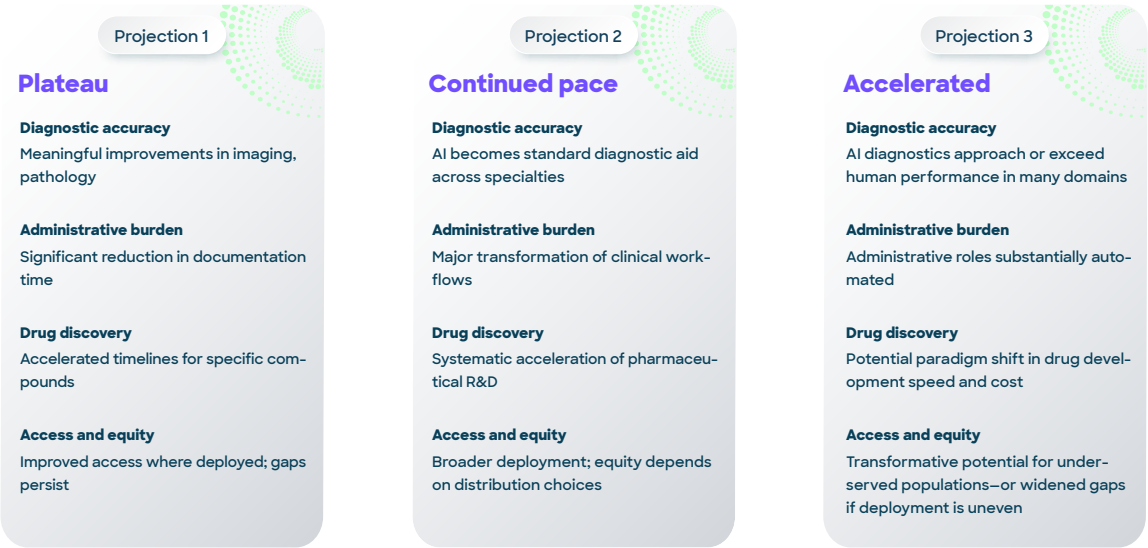
Healthcare represents both enormous potential for AI benefit (diagnostics, drug discovery, administrative efficiency) and significant risks (safety, equity, workforce transformation). The sector's combination of high stakes, data richness, and regulatory complexity makes it a critical test case.

### 3.10.2 Key evidence (2025–2026)

| Finding                                                                                                                 | Source |
|-------------------------------------------------------------------------------------------------------------------------|--------|
| AI triage for intracranial hemorrhage reduced 30-day mortality from 27.7% to 17.5% (37% relative risk reduction).       | [130]  |
| Sepsis AI detected 82% of cases (nearly 2x prior methods) with fewer false alarms, linking to ~20% mortality reduction. | [131]  |
| FDA-cleared AI foundation model achieved 97% sensitivity, 98% specificity across 14 critical CT findings.               | [132]  |
| Over 90% of healthcare leaders prioritise AI for documentation and clinical decision-making within 12–24 months.        | [133]  |
| Generative AI accelerates drug discovery from years to months through molecular generation and interaction simulation.  | [134]  |
| AI healthcare market: \$14.6B (2024), projected \$80–188B by 2030–2036.                                                 | [135]  |

# 3.10.3 Impacts by projection

## First-order impacts



## Second-order impacts



### 3.10.4 What this looks like in each projection

**P1 (Plateau):** A radiologist now works alongside AI that pre-screens imaging for critical findings, allowing faster triage of urgent cases. Documentation time has dropped significantly as ambient AI scribes capture clinical encounters. Drug companies use AI to accelerate certain phases of discovery, but the overall development timeline remains years-long due to clinical trials and regulatory requirements. Patients in well-resourced settings benefit from earlier diagnosis and more efficient care; those in under-resourced settings see smaller improvements. The healthcare workforce remains largely stable, though new roles emerge around AI oversight and maintenance.

**P2 (Continued pace):** AI becomes a standard clinical tool across most specialties. Primary care physicians rely on AI-assisted diagnosis that catches conditions they might have missed; specialists find their expertise augmented but also partially commoditised. Pharmaceutical companies achieve breakthroughs that previously seemed decades away—new treatments for previously intractable conditions reach clinical trials in compressed timeframes. Healthcare workforce discussions intensify: some roles (medical coders, certain diagnostic technicians) face displacement, while others (AI-healthcare integrators, patient navigators) grow. Rural and underserved areas gain access to specialist-level diagnostic capability through telemedicine augmented by AI.

**P3 (Accelerated):** Healthcare transforms more in 3 years than in the previous 30. AI systems achieve diagnostic performance exceeding human specialists across many conditions; the role of physicians shifts towards judgment, communication, and oversight rather than pattern recognition. Drug discovery pipelines produce candidates in months rather than years. Some hospitals restructure dramatically, with AI handling much of the diagnostic and administrative work while humans focus on physical care and patient interaction. Medical education debates what to teach when AI handles much of the cognitive work. Healthcare costs potentially decline substantially—or the benefits are captured by providers rather than patients, depending on policy choices.

### 3.10.5 What if done right

A real system for individual wellbeing emerges, supporting humans in terms of nutrition, sleep, exercise, diagnostics, and treatments. Healthcare shifts from reactive sick-care to proactive wellness optimisation. AI systems that understand individual biology, lifestyle, and preferences provide personalised guidance that actually works—because it's tailored to each person rather than population averages.

This is available for everyone to decide how deeply integrated, which data to share, and how intrusive AI should be in their own life. Privacy and autonomy are preserved: some people choose comprehensive monitoring and proactive intervention; others prefer minimal involvement; most find a middle ground that suits their preferences. The system respects these choices rather than imposing one-size-fits-all approaches.

The result is dramatically improved health outcomes at lower cost, with benefits distributed equitably rather than concentrated among the wealthy. The combination of AI-enabled precision, respect for individual choice, and universal access creates healthcare that actually promotes health rather than just treating disease.

### 3.10.6 Opportunities and risks

#### Opportunities:

- Earlier and more accurate diagnosis saving lives
- Liberation of clinicians from administrative burden
- Accelerated development of new treatments
- Expanded access to specialist-level insights
- Personalized wellness systems respecting individual choice

#### Risks:

- Safety incidents undermining trust
- Equity gaps if deployment favours affluent populations
- Workforce disruption in clinical and administrative roles
- Liability uncertainty slowing beneficial adoption

### 3.11 Impact category 9: Education system

#### 3.11.1 Why this category matters

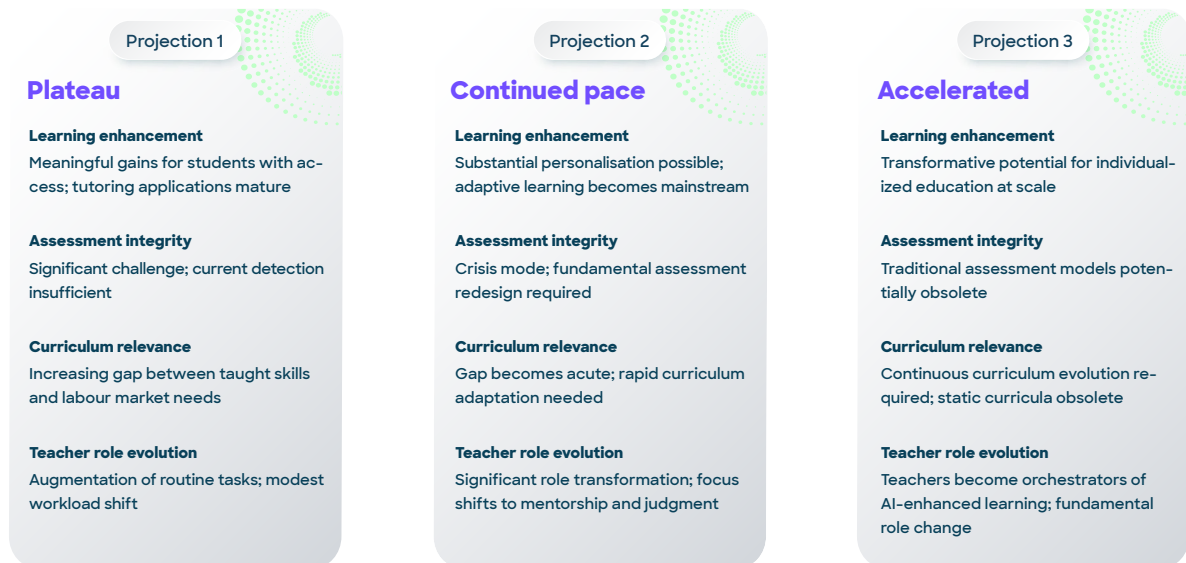
Education systems must simultaneously adapt to AI as a tool, prepare students for AI-transformed labour markets, and maintain integrity in an era of AI-generated content. The challenge spans curriculum, pedagogy, assessment, and equity.

#### 3.11.2 Key evidence (2025–2026)

| Finding                                                                                                               | Source |
|-----------------------------------------------------------------------------------------------------------------------|--------|
| 62% increase in test scores for students using AI-powered instruction due to gap identification and instant feedback. | [136]  |
| 86% of K-12 students used AI in 2024–25, amid concerns about downsides.                                               | [137]  |
| AI use for UK university assessments rose from 53% (2024) to 88% (2025).                                              | [138]  |
| 71% of teachers report added burden verifying student work authenticity.                                              | [139]  |
| 69% of teachers say AI improves methods; 55% report more time for student interaction.                                | [140]  |
| 74% of US districts provided AI training by 2025 (up from 48% in 2023).                                               | [141]  |
| AI encourages cognitive offloading and dependency, weakening foundational knowledge and critical thinking.            | [142]  |

### 3.11.3 Impacts by projection

#### First-order impacts



#### Second-order impacts



### 3.11.4 What this looks like in each projection

**P1 (Plateau):** A university professor struggles with the reality that 80%+ of students use AI for writing assignments. Detection tools are unreliable; redesigning all assessments is time-consuming. Some departments shift towards oral examinations or project-based assessment; others attempt to «ban» AI with limited success. K-12 schools vary widely—some embrace AI tutoring tools that show genuine learning improvements; others lack resources or training to adopt. The gap between well-resourced and under-resourced schools widens. Curriculum discussions intensify about what skills students need when AI handles much of the routine work.

**P2 (Continued pace):** Educational assessment enters genuine crisis. Traditional examinations become nearly meaningless as AI can produce expert-level work on demand. Universities experiment radically—some eliminating written assignments entirely; others developing sophisticated proctored oral or practical examinations. K-12 education splits: leading districts redesign curricula around AI augmentation, teaching students to use AI as a tool while developing human skills like creativity, ethics, and judgment. Trailing districts continue traditional approaches that increasingly disconnect from labour market reality. Teacher roles transform; the best teachers become AI-augmented learning coaches while others struggle with obsolescence.

**P3 (Accelerated):** Education systems face existential questions about their purpose. If AI can perform most cognitive tasks at expert level, what should schools teach? Some argue for emphasis on distinctly human qualities—emotional intelligence, physical skills, ethical reasoning, creativity. Others advocate for intensive AI fluency so students can direct and supervise AI systems. Traditional age-graded, subject-divided schooling looks increasingly anachronistic. Some jurisdictions attempt radical redesigns; others cling to traditional models that produce graduates poorly matched to available work. Educational inequality becomes a dominant political issue as the gap between AI-prepared and AI-unprepared young people creates visible divergence in life outcomes.

### 3.11.5 What if done right

Imagine kids grow up with personal teachers that individually and over long periods of time support them in education. Not replacing human teachers, but augmenting them—AI tutors that know each child’s learning style, interests, current understanding, and optimal challenges, available whenever the child wants to learn.

Highly individualized content based on their interests, their capabilities, and their learning methods means that every child can learn effectively, not just those who happen to fit the standard approach. A child fascinated by dinosaurs learns math through paleontology; a child who struggles with reading gets patient, adaptive support that never judges or gets frustrated.

A unified understanding emerges of what counts in human development: critical thinking, asking the right questions, emotional skills, ethical reasoning, creativity, collaboration. These are the capacities that AI cannot replace and that become more valuable as AI handles routine cognitive tasks.

Education becomes available for everyone independent of parents’ wealth or country of origin. The accident of birth no longer determines access to excellent education. A child in a rural village has access to the same quality of personalised instruction as a child in a wealthy capital city. The combination of AI capability and universal access creates educational equity that seemed impossible.

### 3.11.6 Opportunities and risks

#### Opportunities:

- Personalized learning at scale
- Teachers freed from routine tasks for higher-value interaction
- Democratized access to high-quality educational content
- Real-time feedback and adaptive curricula
- Universal access regardless of geography or wealth

#### Risks:

- Assessment integrity crisis undermining credential value
- Cognitive dependency reducing independent thinking
- Digital divide creating educational inequality
- Curriculum lag leaving graduates unprepared

## 3.12 Impact category 10: Functioning local institutions & liveability

### 3.12.1 Why this category matters

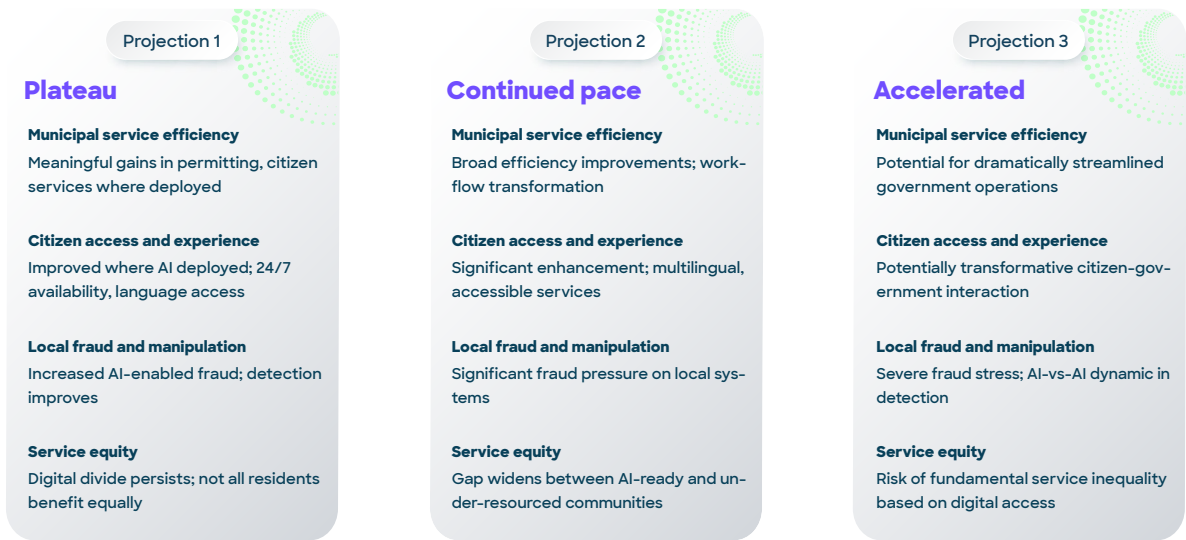
Beyond national-level effects, AI transformation will be experienced most directly at the local level—in municipal services, citizen interactions with government, local economic vitality, and community cohesion. The capacity of local institutions to adapt determines whether AI benefits are broadly shared or concentrated.

### 3.12.2 Key evidence (2025–2026)

| Finding                                                                                                                   | Source |
|---------------------------------------------------------------------------------------------------------------------------|--------|
| 67% of cities are actively integrating AI into operations (top priorities: citizen services, infrastructure, compliance). | [143]  |
| Burlington, VT reduced permit approval from 15 weeks to 5–7 weeks using AI systems.                                       | [144]  |
| San Jose AI Upskilling Programme saved \$50,000 and 10,000–20,000 staff hours in first two cohorts.                       | [145]  |
| Only 6% of local governments place AI in their high-priority category.                                                    | [146]  |
| 77% of Americans distrust both businesses and government to use AI responsibly.                                           | [147]  |
| 45% of residents say digital government services still require substantial improvement despite AI adoption.               | [148]  |
| California Community Colleges estimated ~1/3 of financial aid applicants were fraudulent in 2024; AI detection deployed.  | [149]  |

### 3.12.3 Impacts by projection

#### First-order impacts



#### Second-Order Impacts



### 3.12.4 What this looks like in each projection

**P1 (Plateau):** A citizen applying for a building permit in a forward-thinking city encounters a smooth, AI-assisted process—the chatbot answers questions accurately, documents are processed in days rather than weeks, and translation services make the process accessible. But in a neighboring jurisdiction with less capacity, the experience is unchanged from a decade ago. Some cities use AI to improve pothole detection and infrastructure maintenance; others lack budget or expertise. Fraud targeting local governments increases—AI-generated voice impersonation attempts to redirect payments—but detection tools help. Overall, service quality varies significantly by jurisdiction, creating visible inequality in local government capability.

**P2 (Continued Pace):** The gap between AI-enabled and traditional local government widens dramatically. Leading cities offer genuinely transformed citizen experience—near-instant permitting, proactive service delivery, multilingual 24/7 assistance. Lagging jurisdictions see resident frustration as expectations set by leading cities aren't met locally. Local government employment shifts as AI handles routine administrative work; some jurisdictions manage transitions smoothly while others face workforce conflicts. Fraud pressure intensifies—sophisticated AI-enabled schemes target benefit programmes, tax systems, and emergency services. Communities that invested early in AI capacity thrive; those that didn't struggle to maintain basic services.

**P3 (Accelerated):** Local government faces potential crisis. The speed of change exceeds institutional capacity in many jurisdictions. Cities that successfully transformed deliver services at quality and efficiency levels that seemed impossible—but they're the minority. Other communities face simultaneous challenges: increased demand from residents affected by labour market disruption, sophisticated fraud depleting resources, and inability to compete for talent with private sector. Some small cities and rural areas find that AI enables service delivery they previously couldn't afford; others are left further behind. Regional inequality becomes a dominant political issue as the local experience of AI transformation varies enormously by geography. Groups of people demand slowing down change, which is not possible. New movements and communities emerge that demand an AI-free life, seeking to preserve traditional ways of living.

### 3.12.5 What if done right

Due to a broad diffusion of AI-powered services, AI is broadly considered as benefiting communities and individual lives. The transformation is not imposed from above but emerges from genuine local benefit: services that actually work better, costs that actually decrease, access that actually expands.

Resistance to change and even movements that want to reverse to a life before AI appear and are accepted. Societies make space for communities and individuals who choose different relationships with technology. This diversity is respected rather than suppressed.

But due to the overall benefits in local communities, the acceptance of AI remains strong. When citizens experience AI as making their lives better—faster services, fairer treatment, more responsive government—the political foundation for continued adoption remains solid. Trust is earned through demonstrated benefit, not demanded through mandate.

The result is local communities that are more liveable, more equitable, and more resilient—not despite AI but because AI is deployed in service of community values rather than as an end in itself.

## 3.12.6 Opportunities and risks

### Opportunities:

- More efficient, accessible, responsive local government
- 24/7 citizen services in multiple languages
- Data-driven resource allocation improving outcomes
- Liberation of local workers from routine tasks
- Services that benefit all community members

### Risks:

- Digital divide creating two-tier local citizenship
- Institutional capacity gaps in under-resourced communities
- Trust erosion from AI failures or opaque decision-making
- Local government unable to adapt fast enough to maintain services
- Social fragmentation from pace of change
- 3.13 Cross-cutting observations

## 3.13.1 Patterns across impact categories

### Several patterns emerge across all ten impact categories:

1. Adoption capacity is universally important: In all projections, the ability to adopt, integrate, and govern AI systems is a primary determinant of outcomes. Organisations and communities with higher capacity realise benefits; those without are left behind or harmed.
2. Early-career and vulnerable populations face disproportionate risk: Entry-level workers, digitally excluded populations, and under-resourced institutions bear the earliest and most severe negative impacts.
3. Trust and integrity are fragile: Across information, science, and governance, AI capabilities stress verification and trust systems. Failures here have cascading effects.
4. The gap widens with capability speed: In P3, all impacts intensify—both opportunities and risks. The window for orderly adaptation shrinks.
5. Second-order effects are substantial: Direct AI impacts cascade through systems (labour → tax base → public services → local institutions). Planning must account for these chains.
6. Robotics is now economically transformative: With 99%+ success rates in production and ROI of 1,400–2,070%, physical automation is no longer speculative—it's a near-term competitive imperative.
7. Innovation economics are restructuring: Few-person unicorns, rapid IP depreciation, and collapsing competitive moats require new frameworks for economic policy.

8. Social safety net adequacy varies dramatically: Countries with robust systems can buffer disruption; those without face severe social strain.
9. “Done right” visions are achievable: Each impact category has a positive path—but realising it requires proactive policy, institutional adaptation, and deliberate choices about how AI benefits are distributed.

### 3.13.2 Time-critical impact patterns

Some impacts require attention regardless of projection probability because they are already materialising or have long lead times for response:

| Impact Pattern                         | Why time-critical                                                          | Projections where Acute |
|----------------------------------------|----------------------------------------------------------------------------|-------------------------|
| Entry-level labour displacement        | Already visible (6–20% in exposed roles); career effects compound          | All (P1–P3)             |
| Software engineering transformation    | SWE-bench at human parity; majority of code AI-written at leading firms    | All (P1–P3)             |
| Assessment integrity in education      | 88% of UK university students using AI; crisis already emerging            | All (P1–P3)             |
| Information environment degradation    | 52% machine-generated content; detection lagging                           | All (P1–P3)             |
| SME adoption gap                       | Structural barrier; widens without intervention                            | P2, P3                  |
| Robotics competitiveness               | Chinese manufacturers at 39% global share; costs declining 20–30% annually | P2, P3                  |
| Local government capacity              | Only 6% prioritise AI; adaptation takes years                              | P2, P3                  |
| Scientific integrity                   | Hallucinations in peer-reviewed venues; trust erosion underway             | All (P1–P3)             |
| Startup economics transformation       | Few-person unicorns emerging; traditional models under pressure            | P2, P3                  |
| Market repricing of AI-exposed sectors | „AI Scare Trade“ demonstrates volatility; structural shifts underway       | P2, P3                  |

### 3.13.3 No-regret impact patterns

Some impact patterns require action regardless of which projection materializes, because they are already emerging (probability = 1 for some level of impact):

- Labour market transition support – needed in all projections; scale varies
- AI literacy and workforce development – essential regardless of capability speed
- Information integrity infrastructure – already urgent; will only intensify
- Trustworthy AI governance capacity – foundational for all other responses
- Digital equity for public services – determines who benefits from AI-enabled government
- Innovation ecosystem adaptation – necessary to remain competitive in any projection

These no-regret patterns inform the measures identified in Chapter 4.

### 3.14 Summary: Impact Severity by Projection

| Impact category                 | P1: Plateau                      | P2: Continued pace                         | P3: Accelerated                    |
|---------------------------------|----------------------------------|--------------------------------------------|------------------------------------|
| Labour market & skills          | Moderate (entry-level pressure)  | High (broad displacement)                  | Severe (structural transformation) |
| Public finance & social systems | Manageable                       | Significant stress                         | Potential crisis                   |
| Industry & competitiveness      | Moderate (adoption gap)          | High (competitive gap + market volatility) | Severe (structural decline risk)   |
| Innovation & startups           | Moderate (efficiency pressure)   | High (model disruption)                    | Severe (fundamental restructuring) |
| Science system                  | Mixed (acceleration + integrity) | High (integrity crisis)                    | Severe (fundamental challenges)    |
| Security & resilience           | Elevated                         | High                                       | Severe (systemic risk)             |
| Digital public sphere           | High (integrity)                 | Severe (trust crisis)                      | Potentially catastrophic           |
| Health & care                   | Positive (efficiency)            | Strongly positive (if governed)            | Transformative (with governance)   |

|                    |                            |                              |                                    |
|--------------------|----------------------------|------------------------------|------------------------------------|
| Education system   | Significant (integrity)    | High (integrity + relevance) | Severe (systemic re-design needed) |
| Local institutions | Moderate (uneven adoption) | High (capacity gaps)         | Severe (institutional overload)    |

### 3.15 What this chapter does not determine

This chapter maps impacts but does not:

1. Prescribe specific measures – these are derived in Chapter 4 based on impact severity and probability.
2. Assign responsibility – the owner ecosystem for each measure is specified in Chapter 4.
3. Evaluate European response capacity – this informs measure feasibility in Chapter 4.
4. Prioritize among impacts – prioritisation requires combining impact severity with projection probability (Chapter 4).
- 5.

# Chapter 4:

## Measures & appliedAI portfolio logic

### 4.1 Purpose of this chapter

This chapter translates the impacts identified in Chapter 3 into concrete measures, distinguishing between:

- Part A: What should be addressed overall (the full measures inventory, regardless of who implements)
- Part B: What appliedAI Institute will do (our portfolio selection based on purpose fit and feasibility)

**The logic:**

1. Start from impact items in Chapter 3
2. Weight by probability (from Chapter 2) and time criticality
3. Identify measures that address these impacts
4. Distinguish no-regret measures (needed in all projections) from projection-conditional measures
5. Select appliedAI's portfolio based on purpose fit and feasibility

### 4.2 Measures design principles

#### 4.2.1 Derivation from Impacts

**Every measure must map back to:**

- An impact (Chapter 3)
- A probability basis (projection probabilities from Chapter 2 and/or measure-level probability)
- Time criticality (0–36 months)

## 4.2.2 Measure types

| Type                   | Definition                                                                                                               | Probability basis                 |
|------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| No-regret              | Needed regardless of which projection materializes; addresses impacts already emerging or foundational for all responses | Probability = 1                   |
| Projection-conditional | Becomes critical primarily in P2 or P3; may be premature or lower priority if P1 materializes                            | Probability = projection-weighted |

## 4.2.3 Scoring dimensions

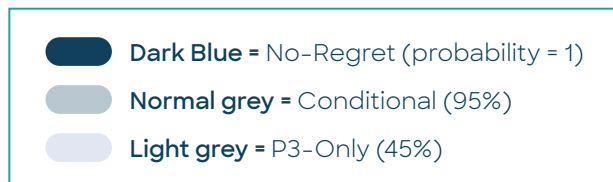
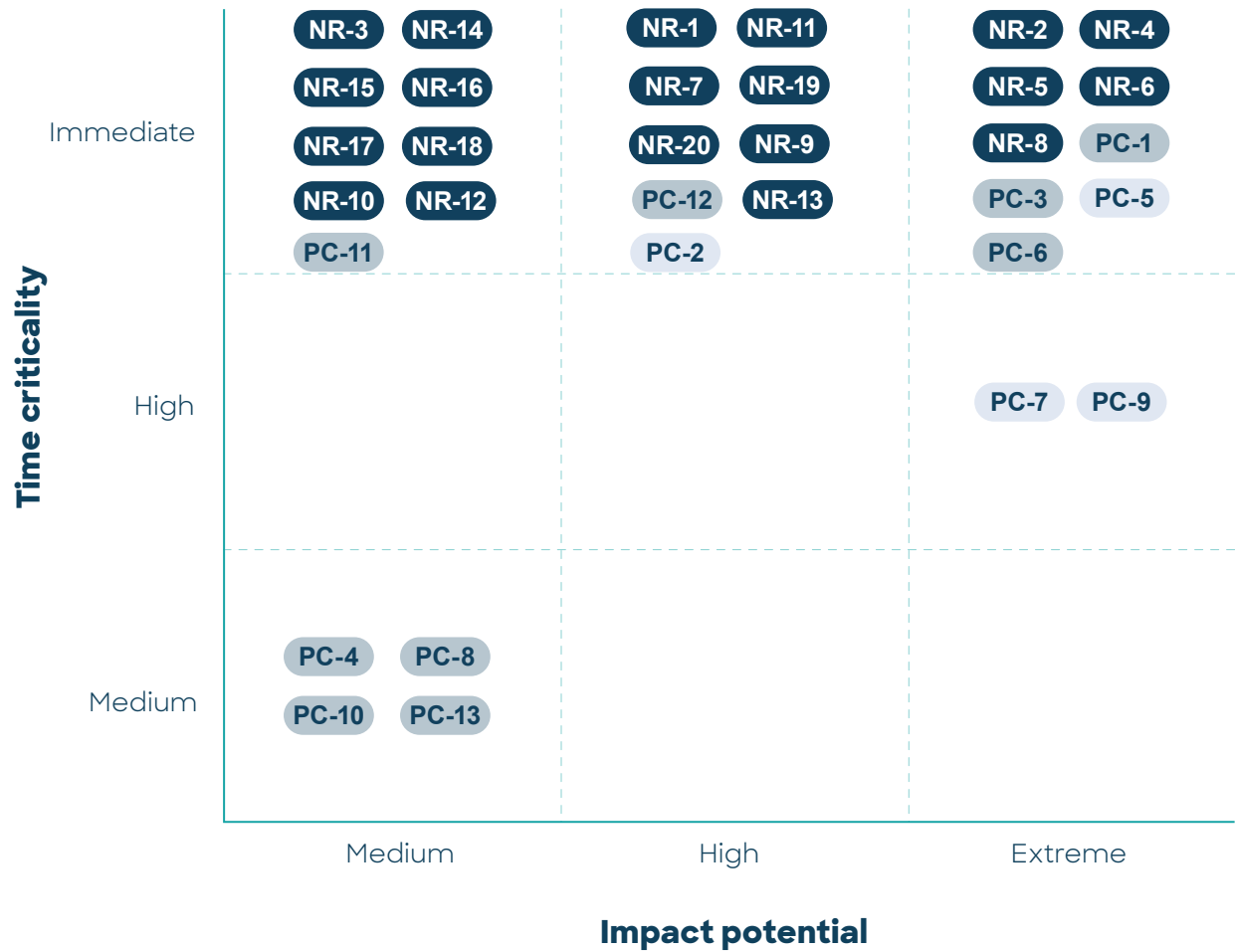
| Dimension            | Definition                                                                                        |
|----------------------|---------------------------------------------------------------------------------------------------|
| Impact potential     | Scale and severity of the impact addressed (low / medium / high / extreme)                        |
| Probability          | Likelihood that the measure will be needed (1 for no-regret; projection-weighted for conditional) |
| Time criticality     | Urgency of action (low / medium / high) within 0–36 months                                        |
| Overall value signal | Qualitative synthesis of the above                                                                |

## 4.3 Part A – measures inventory (what should be addressed overall)

This section presents the full inventory of measures derived from Chapter 3 impacts. These measures represent what Europe and its institutions should address—regardless of which specific organisation implements them.

### 4.3.1 No-regret measures (probability = 1)

These measures are valuable across all projections and should be pursued immediately:



NR-1

## Development of European target images for people in the AI age

### Linked impact (s)

Labour, Education, Social Systems

### Evidence basis

Public discourse fragmented; no coherent European narrative

### Intended effect

Establish shared vision for human flourishing in AI-transformed society; guide policy and investment

### Suggested owners

EU institutions, national governments, civil society, research institutions

NR-7

## Promotion and targeted settlement of international AI talent in Europe

### Linked impact (s)

Industry, Science, Competitiveness

### Evidence basis

Talent flowing to US; European retention challenging

### Intended effect

Address talent shortage; reduce brain drain; build European AI capacity

### Suggested owners

National governments, universities, industry

NR-2

## Establishment of genuine technical and regulatory AI competence centres at the highest level

### Linked impact (s)

All categories

### Evidence basis

Only 6% of local governments prioritise AI; regulatory capacity strained; AI Office currently lags behind tech developments by >1 year; similar gaps at national level

### Intended effect

Build institutional capacity for AI governance across government; ensure informed decision-making

### Suggested owners

National governments, EU institutions, public sector academies

NR-8

## Supporting and scaling promising European AI start-ups (including government contracts)

### Linked impact (s)

Industry, Competitiveness, Innovation

### Evidence basis

EU AI investment 7% of US levels; ecosystem underdeveloped

### Intended effect

Build European AI ecosystem; reduce dependency; create jobs

### Suggested owners

National governments, EU institutions, procurement bodies

NR-3

## Creation of concrete 2-year action plan addressing all three scenarios with continuous monitoring

### Linked impact (s)

All categories

### Evidence basis

Current planning assumes single trajectory; lacks contingency

### Intended effect

Ensure preparedness across scenarios; enable rapid response to trajectory changes

### Suggested owners

National governments, EU Commission, coordinating bodies

NR-9

## Conceptualization of taxation of agentic value creation

### Linked impact (s)

Public Finance, Labour

### Evidence basis

Current systems assume labour-based value creation; AI shifting value creation patterns

### Intended effect

Develop frameworks for taxing AI-generated value; ensure fiscal sustainability as labour tax base erodes

### Suggested owners

Finance ministries, tax policy institutions, research institutions

NR-4

## Introduction of mandatory AI monitoring (similar to UK AI Safety Taskforce)

### Linked impact (s)

Security, Industry, All

### Evidence basis

Current monitoring fragmented; lags capability advancement

### Intended effect

Track AI capability advancement; identify risks early; enable evidence-based policy

### Suggested owners

EU AI Office, national AI authorities, research institutions

NR-10

## Creation of government-supported individual teacher platform for future of education

### Linked impact (s)

Education, Labour, Social Systems

### Evidence basis

88% UK students using AI; traditional education models under stress; need for scalable personalised learning

### Intended effect

Enable personalised AI-assisted education at scale; prepare workforce for AI transformation

### Suggested owners

Education ministries, national governments, EdTech sector

NR-5

## Systematic unblocking of delaying mechanisms for public procurement

### Linked impact (s)

All categories

### Evidence basis

Procurement 12-24 months; AI advancing in weeks/months; bureaucratic delays prevent adoption

### Intended effect

Close gap between AI advancement speed and institutional adaptation; enable rapid deployment of AI solutions

### Suggested owners

All levels of government, regulatory bodies, procurement offices

NR-11

## Development and introduction of verifiable digital identity for people on the open Internet

### Linked impact (s)

Digital Public Sphere, Security, Democracy

### Evidence basis

52% content now machine-generated; verification impossible

### Intended effect

Distinguish humans from AI actors; combat manipulation; preserve trust

### Suggested owners

EU institutions, national governments, standards bodies

NR-6

## Moonshots for next-generation AI architectures from Europe

### Linked impact (s)

Industry, Science, Competitiveness, Sovereignty

### Evidence basis

EU controls 4.8-5% of compute; dependent on foreign models; Nested Learning, Causal AI emerging as new paradigms

### Intended effect

Develop European capacity in frontier AI architectures; reduce dependency on foreign models; pursue paradigm-shifting research

### Suggested owners

EU institutions, research institutions, national governments

NR-12

## Monitoring and rapid response systems for massive digital manipulation attacks

### Linked impact (s)

Digital Public Sphere, Democracy, Security

### Evidence basis

Deepfakes doubling every 6 months; phishing up 400%; agentic attacks emerging

### Intended effect

Detect and respond to AI-enabled influence operations; protect elections; enable coordinated response to manipulation waves

### Suggested owners

National security agencies, election authorities, platforms

NR-13

### Skills exposure study (Europe-wide, role/task-based)

**Linked impact (s)**

Labour Market, Public Finance

**Intended effect**

Identify most affected roles and regions; enable targeted response

**Evidence basis**

Entry-level displacement 6–20% already visible

**Suggested owners**

Research institutions, labour ministries, appliedAI

NR-17

### Assessment redesign frameworks for education

**Linked impact (s)**

Education

**Intended effect**

Address integrity crisis; maintain credential value

**Evidence basis**

88% UK students using AI; traditional assessment failing

**Suggested owners**

Education ministries, examination boards, universities

NR-14

### Scalable reskilling and upskilling platform (role-based)

**Linked impact (s)**

Labour Market, Industry

**Intended effect**

Enable workforce transition; maintain employability; scale AI literacy across workforce

**Evidence basis**

Current training capacity insufficient for projected need; 13.48% EU adoption; skills shortage primary barrier

**Suggested owners**

Education providers, industry, public employment services, EU institutions

NR-18

### Scientific integrity infrastructure for AI era

**Linked impact (s)**

Science, Innovation

**Intended effect**

Address hallucination and fraud risks; maintain trust in research

**Evidence basis**

53 NeurIPS papers with hallucinated citations; fraud scaling

**Suggested owners**

Research funders, publishers, scientific academies

NR-15

### Trustworthy AI engineering playbooks and reference architectures

**Linked impact (s)**

All categories (enabler)

**Intended effect**

Establish practical standards for responsible deployment

**Evidence basis**

Adoption barriers include uncertainty about „how to do it right“

**Suggested owners**

Standards bodies, industry associations, appliedAI

NR-19

### Innovation ecosystem adaptation framework

**Linked impact (s)**

Innovation, Startups, Competitiveness

**Intended effect**

Support European startups in AI-transformed competitive landscape; address few-person unicorn dynamics

**Evidence basis**

Traditional startup models under pressure; revenue per employee transforming

**Suggested owners**

Innovation agencies, VCs, startup associations, appliedAI

NR-16

### Local government adoption blueprints (municipal workflows)

**Linked impact (s)**

Local Institutions, Liveability

**Intended effect**

Enable local government AI adoption; reduce capacity barriers

**Evidence basis**

67% of cities integrating AI but most lack blueprints

**Suggested owners**

Municipal associations, national governments, appliedAI

NR-20

### SME onboarding support for AI adoption

**Linked impact (s)**

Industry, Competitiveness, Local Institutions

**Intended effect**

Enable SMEs to adopt AI; close adoption gap between large enterprises and smaller firms

**Evidence basis**

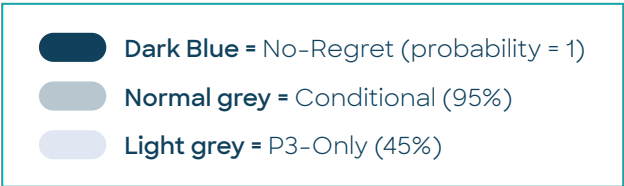
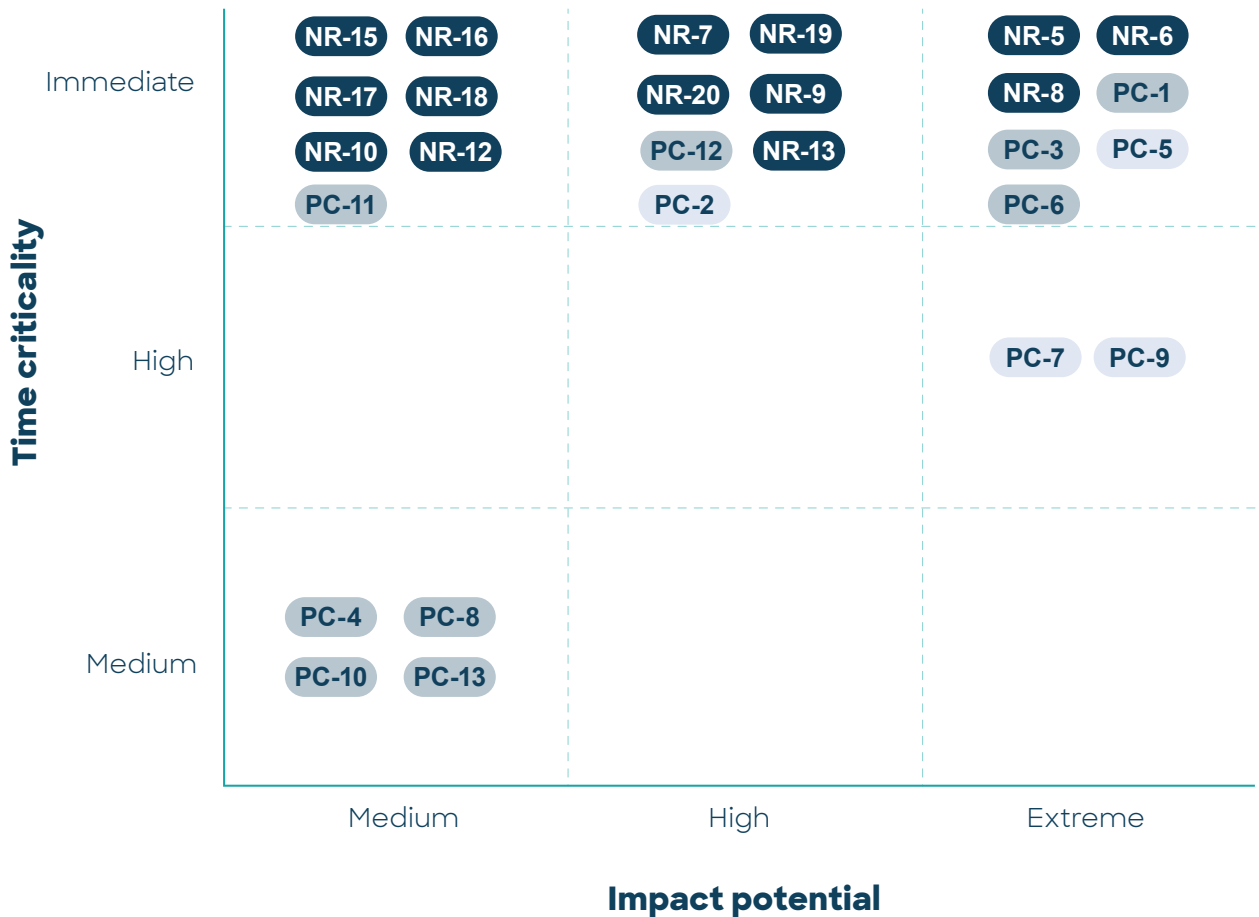
41% of large EU enterprises have AI vs. ~12% of SMEs; SMEs lack resources and expertise for adoption

**Suggested owners**

Industry associations, chambers of commerce, appliedAI, regional development agencies

### 4.3.2 Projection-conditional measures (P2/P3)

These measures become critical primarily if P2 or P3 materializes. Given our probability estimates (P2: 50%, P3: 45%), these warrant serious preparation, while P2 measures become relevant for P2 and P3 giving a 95% total probability that those measures become relevant:



PC-1

### Establishment of European early warning system for technological leaps, misuse risks, and geopolitical shifts

**Linked impact (s)**  
All categories

**Probability basis**  
S2/S3 (95%)

**Intended effect**

Detect and respond to rapid capability changes; maintain strategic awareness

**Suggested owners**

Intelligence services, EU AI Office, research institutions

PC-8

### Healthcare AI governance fast-track frameworks

**Linked impact (s)**  
Health & Care

**Probability basis**  
S2/S3 (95%)

**Intended effect**

Enable rapid deployment while maintaining safety

**Suggested owners**

Health ministries, regulatory agencies

PC-2

### Establishment of verified and decoupled „economic internet“ for strategically important entities

**Linked impact (s)**  
Security, Industry, Critical Infrastructure

**Probability basis**  
Primarily S3 (45%)

**Intended effect**

Ensure communication resilience; protect against AI-enabled attacks

**Suggested owners**

National security agencies, critical infrastructure operators

PC-9

### Emergency fiscal measures for tax base transition

**Linked impact (s)**  
Public Finance

**Probability basis**  
Primarily S3 (45%)

**Intended effect**

Prepare revenue alternatives if labour taxes decline rapidly

**Suggested owners**

Finance ministries

PC-3

### Preparation for AI agent-conducted cyberattacks on undetected vulnerabilities

**Linked impact (s)**  
Security, Industry

**Probability basis**  
S2/S3 (95%)

**Intended effect**

Develop defensive capabilities against autonomous attack systems

**Suggested owners**

Cybersecurity agencies, critical infrastructure operators

PC-10

### AI native educational pathways (parallel to traditional)

**Linked impact (s)**  
Education

**Probability basis**  
S2/S3 (95%)

**Intended effect**

Develop education models for AI-transformed economy

**Suggested owners**

Education ministries, universities

PC-4

### Strategic massive expansion of European computing capacities and energy supply

**Linked impact (s)**  
Industry, Competitiveness, Sovereignty

**Probability basis**  
S2/S3 (95%)

**Intended effect**

Reduce dependency; enable European AI development

**Suggested owners**

National governments, EU institutions, energy providers

PC-11

### International coordination on AI safety and governance

**Linked impact (s)**  
Security, All categories

**Probability basis**  
S2/S3 (95%)

**Intended effect**

Align responses across jurisdictions; prevent race to bottom

**Suggested owners**

Foreign ministries, EU institutions, international bodies

PC-5

### Establishment of rapid intervention mechanisms for disruptive AI leaps

**Linked impact (s)**  
All categories

**Probability basis**  
Primarily S3 (45%)

**Intended effect**

Enable fast policy response to unexpected capability jumps

**Suggested owners**

National governments, EU institutions

PC-12

### Union-Industry agreements on joint visions for the future of work

**Linked impact (s)**  
Labour, Industry, Social Cohesion

**Probability basis**  
S2/S3 (95%)

**Intended effect**

Create shared understanding between employers and workers on AI-driven workplace transformation; enable collaborative transition planning

**Suggested owners**

Trade unions, industry associations, labour ministries, social partners

PC-6

### Greenfield AI-First test environments (education, health, food, energy, materials, social)

**Linked impact (s)**  
Multiple sectors

**Probability basis**  
S2/S3 (95%)

**Intended effect**

Develop and test AI-native approaches without legacy constraints

**Suggested owners**

National governments, sector agencies, research institutions

PC-13

### Simulation of new social systems

**Linked impact (s)**  
Public Finance, Labour, Social Systems

**Probability basis**  
Primarily S3 (45%)

**Intended effect**

Model and test alternative social contracts (e.g., UBI variants, new welfare models) before implementation becomes urgent

**Suggested owners**

Social policy institutions, research institutions, finance ministries

PC-7

### Contingency concepts for severe labour displacement

**Linked impact (s)**  
Labour, Public Finance, Social Cohesion

**Probability basis**  
Primarily S3 (45%)

**Intended effect**

Prepare response options for rapid workforce disruption

**Suggested owners**

Labour ministries, social policy institutions

### 4.3.3 Measures prioritisation matrix

The following matrix visualizes measure priority based on probability and impact potential:

|                                | Low im-<br>pact | Medium impact                              | High impact                                               | Extreme im-<br>pact              |
|--------------------------------|-----------------|--------------------------------------------|-----------------------------------------------------------|----------------------------------|
| Probability = 1<br>(No-regret) |                 | NR-3, NR-14, NR-15,<br>NR-16, NR-17, NR-18 | NR-2, NR-4, NR-7, NR-<br>9, NR-10, NR-11, NR-13,<br>NR-19 | NR-1, NR-5, NR-6,<br>NR-8, NR-12 |
| Probability = 95%<br>(P2+P3)   |                 | PC-11                                      | PC-4, PC-8, PC-10, PC-<br>12, PC-13                       | PC-1, PC-3, PC-6                 |
| Probability = 45%<br>(P3 only) |                 |                                            | PC-2                                                      | PC-5, PC-7,<br>PC-9              |

## 4.4 Part B – appliedAI Portfolio Selection (What We Will Do)

### 4.4.1 Portfolio selection filters

appliedAI Institute selects its portfolio from the measures inventory based on four filters:

| Filter                         | Definition                                                                                                                                                 | Implications                                                                                               |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Purpose fit                    | Alignment with appliedAI's mission: em-<br>powering professionals to develop and<br>apply AI technologies in a trustworthy<br>manner                       | Focus on competence build-<br>ing, responsible deployment,<br>practical guidance                           |
| Feasibility                    | Within appliedAI's mandate and capabili-<br>ties as a non-profit research and compe-<br>tence organisation                                                 | Cannot directly change polit-<br>ical frameworks; can provide<br>evidence, methods, training,<br>convening |
| Comparative ad-<br>vantage     | Areas where appliedAI has distinctive<br>capability: methods development, engi-<br>neering expertise, ecosystem conven-<br>ing, neutral guidance, training | Focus on areas where we can<br>add unique value vs. others                                                 |
| Ecosystem comple-<br>mentarity | Avoid duplication; partner where others<br>lead                                                                                                            | Focus on gaps; enable rather<br>than replace other actors                                                  |

## 4.4.2 What appliedAI cannot do directly

To be transparent about scope, appliedAI Institute cannot directly:

- Change political frameworks or legislation
- Provide fiscal resources at government scale
- Build physical infrastructure (compute, energy)
- Regulate or enforce compliance
- Make binding policy decisions

However, we can:

- Provide evidence and analysis to inform decisions
- Develop methods, frameworks, and playbooks
- Train professionals across sectors
- Convene stakeholders and facilitate collaboration
- Create reference implementations and tools
- Highlight European solutions and capabilities
- Build communities of practice

## 4.4.3 appliedAI opportunity fields

Based on the filters above, we organise our portfolio into four opportunity fields:

Opportunity Field 1: Skills & Workforce Transformation

Purpose: Build measurable AI competence across European professionals

Linked measures: NR-13, NR-14

What we do:

- appliedAI Skills Framework: Standardized competency definitions for AI roles across sectors
- appliedAI Skills Academy (NR-14): Scalable training programmes for professionals in industry and public sector, based on the appliedAI Skills Framework
- Skills Exposure Research (NR-13): Evidence-based analysis of role transformation and reskilling needs, based on the appliedAI Skills Framework
- Target groups: Corporate professionals, public sector employees, executives, educators

- Opportunity Field 2: Trustworthy AI Engineering & Adoption
- Purpose: Enable responsible, effective AI deployment through practical methods and engineering support
- Linked measures: NR-2, NR-5, NR-15, NR-19

#### What we do:

- AI Act Accelerator (NR-2 support, NR-5, NR-15): Together with the Bavarian Ministry of Digital Affairs, includes:
- AI Act Implementation Support: Practical guidance for compliance
- Engineering Playbooks: Practical guidance for trustworthy AI system development
- Agent-First Operations (NR-19): Coordinating and contributing to open-source AI-first functional blueprints with a sovereign tech stack

Target groups: AI engineers, product teams, compliance officers, CTOs

Opportunity Field 3: Local Implementation & Ecosystem Building

Purpose: Enable AI adoption at local level and strengthen European AI ecosystem

Linked measures: NR-8, NR-16, NR-19, NR-20

#### What we do:

- Municipal Agent-First Blueprints (NR-16): Practical guides for local government AI adoption with agentic workflows
- SME Onboarding Support (NR-20): Enabling smaller organisations to benefit from AI through structured adoption programmes
- Startup-SME Matching (NR-20, NR-8): Connecting European startups with SME use cases based on Startup and Use Case Landscapes

Target groups: Local governments, regional development agencies, SMEs, ecosystem builders, startups

Opportunity Field 4: Policy Handrails & Decision Support

Purpose: Provide evidence and guidance for AI governance and policy

Linked measures: NR-1, NR-3, SC-1

#### What we do:

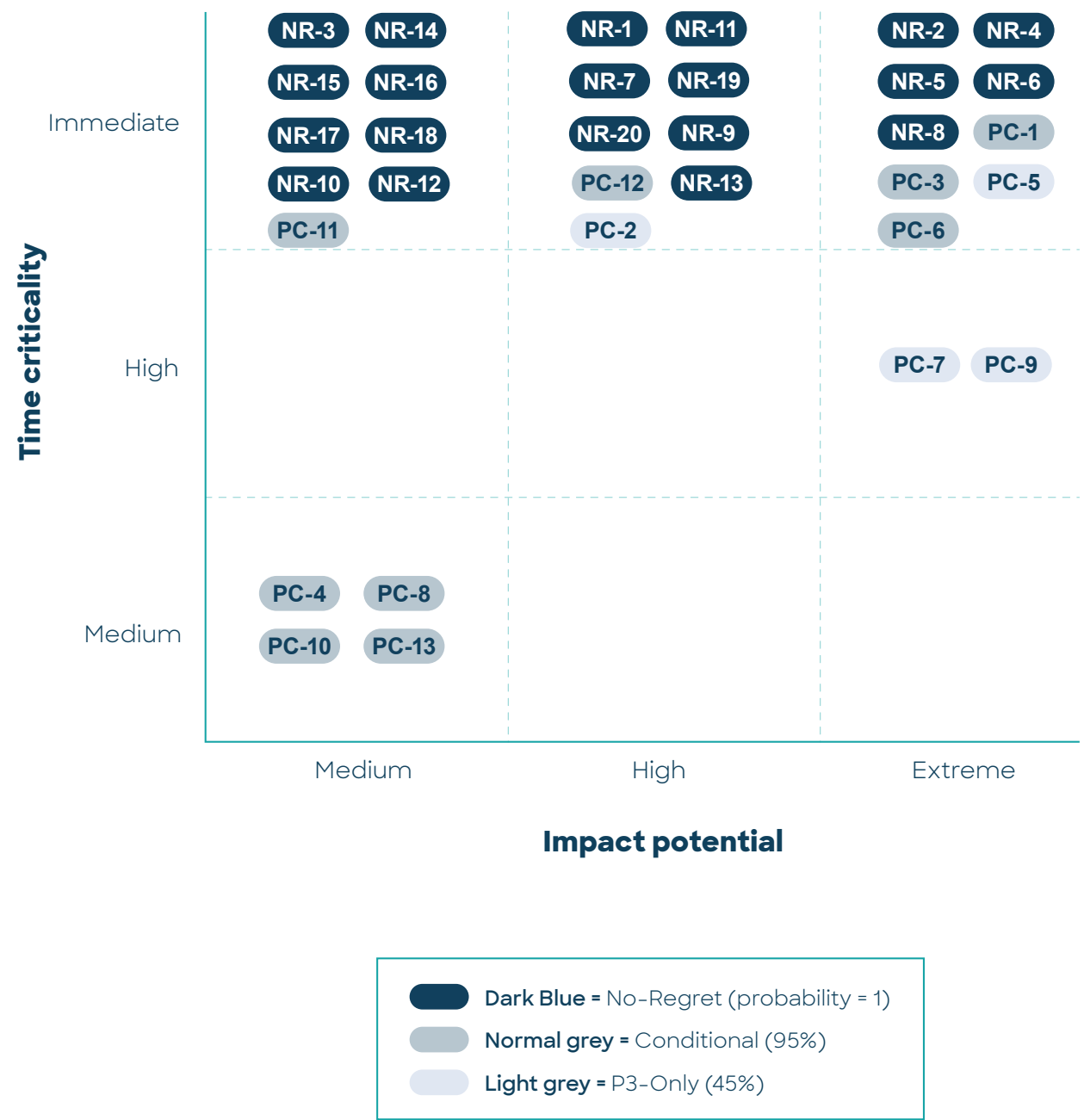
- AI Projection Whitepaper (NR-1, NR-3, SC-1 support): Annual strategic orientation with continuous updates—this document
- Policy Briefs: Concise guidance on specific AI governance challenges
- Target groups: Policy makers, regulators, international bodies

#### 4.4.4 Portfolio summary and measure mapping

| Opportunity Field       | Primary measures addressed | appliedAI role                                     | Key outputs                                                       |
|-------------------------|----------------------------|----------------------------------------------------|-------------------------------------------------------------------|
| Skills & Workforce      | NR-13, NR-14               | Lead on frameworks, training, research             | Skills Framework, Academy, Exposure Studies                       |
| Trustworthy Engineering | NR-2, NR-5, NR-15, NR-19   | Create methods, build references, provide guidance | AI Act Accelerator, Engineering Playbooks, Agent-First Blueprints |
| Local Implementation    | NR-8, NR-16, NR-19, NR-20  | Enable adoption, convene ecosystems                | Municipal Blueprints, SME Programs, Startup-SME Matching          |
| Policy Handrails        | NR-1, NR-3, PC-1           | Provide evidence, develop guidance                 | AI Projections, Policy Briefs                                     |

Measure-to-portfolio mapping:

The following table shows how the measures inventory maps to appliedAI’s portfolio and external recommendations:



## No-regret measures

|                                                                                                                                                                                                          |                                     |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------|-------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------|---------------------------------------------------|----------------------------------|
| <b>NR-1</b> <b>European target images for AI age</b> <table> <tr> <td>Opportunity field</td> <td>Role</td> </tr> <tr> <td>Policy handrails</td> <td>Input</td> </tr> </table>                            | Opportunity field                   | Role | Policy handrails        | Input                               | <b>NR-14</b> <b>Reskilling platform</b> <table> <tr> <td>Opportunity field</td> <td>Role</td> </tr> <tr> <td>Skills &amp; workforce</td> <td>Lead</td> </tr> </table>                                               | Opportunity field | Role | Skills & workforce                                | Lead                             |
| Opportunity field                                                                                                                                                                                        | Role                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Policy handrails                                                                                                                                                                                         | Input                               |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Opportunity field                                                                                                                                                                                        | Role                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Skills & workforce                                                                                                                                                                                       | Lead                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| <b>NR-2</b> <b>AI competence centres</b> <table> <tr> <td>Opportunity field</td> <td>Role</td> </tr> <tr> <td>Trustworthy engineering</td> <td>Support<br/>(via AI Act Accelerator)</td> </tr> </table>  | Opportunity field                   | Role | Trustworthy engineering | Support<br>(via AI Act Accelerator) | <b>NR-15</b> <b>Engineering playbooks</b> <table> <tr> <td>Opportunity field</td> <td>Role</td> </tr> <tr> <td>Trustworthy engineering</td> <td>Lead<br/>(via AI Act Accelerator)</td> </tr> </table>               | Opportunity field | Role | Trustworthy engineering                           | Lead<br>(via AI Act Accelerator) |
| Opportunity field                                                                                                                                                                                        | Role                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Trustworthy engineering                                                                                                                                                                                  | Support<br>(via AI Act Accelerator) |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Opportunity field                                                                                                                                                                                        | Role                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Trustworthy engineering                                                                                                                                                                                  | Lead<br>(via AI Act Accelerator)    |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| <b>NR-3</b> <b>2-year action plan</b> <table> <tr> <td>Opportunity field</td> <td>Role</td> </tr> <tr> <td>Policy handrails</td> <td>Input</td> </tr> </table>                                           | Opportunity field                   | Role | Policy handrails        | Input                               | <b>NR-16</b> <b>Municipal blueprints</b> <table> <tr> <td>Opportunity field</td> <td>Role</td> </tr> <tr> <td>Local implementation</td> <td>Lead</td> </tr> </table>                                                | Opportunity field | Role | Local implementation                              | Lead                             |
| Opportunity field                                                                                                                                                                                        | Role                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Policy handrails                                                                                                                                                                                         | Input                               |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Opportunity field                                                                                                                                                                                        | Role                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Local implementation                                                                                                                                                                                     | Lead                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| <b>NR-5</b> <b>Procurement unblocking</b> <table> <tr> <td>Opportunity field</td> <td>Role</td> </tr> <tr> <td>Trustworthy engineering</td> <td>Support<br/>(via AI Act Accelerator)</td> </tr> </table> | Opportunity field                   | Role | Trustworthy engineering | Support<br>(via AI Act Accelerator) | <b>NR-19</b> <b>Innovation ecosystem adaptation</b> <table> <tr> <td>Opportunity field</td> <td>Role</td> </tr> <tr> <td>Trustworthy engineering<br/>+ local implementation</td> <td>Contribute</td> </tr> </table> | Opportunity field | Role | Trustworthy engineering<br>+ local implementation | Contribute                       |
| Opportunity field                                                                                                                                                                                        | Role                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Trustworthy engineering                                                                                                                                                                                  | Support<br>(via AI Act Accelerator) |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Opportunity field                                                                                                                                                                                        | Role                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Trustworthy engineering<br>+ local implementation                                                                                                                                                        | Contribute                          |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| <b>NR-8</b> <b>Startup support</b> <table> <tr> <td>Opportunity field</td> <td>Role</td> </tr> <tr> <td>Local implementation</td> <td>Support</td> </tr> </table>                                        | Opportunity field                   | Role | Local implementation    | Support                             | <b>NR-20</b> <b>SME onboarding support</b> <table> <tr> <td>Opportunity field</td> <td>Role</td> </tr> <tr> <td>Local implementation</td> <td>Lead</td> </tr> </table>                                              | Opportunity field | Role | Local implementation                              | Lead                             |
| Opportunity field                                                                                                                                                                                        | Role                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Local implementation                                                                                                                                                                                     | Support                             |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Opportunity field                                                                                                                                                                                        | Role                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Local implementation                                                                                                                                                                                     | Lead                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| <b>NR-13</b> <b>Skills exposure study</b> <table> <tr> <td>Opportunity field</td> <td>Role</td> </tr> <tr> <td>Skills &amp; workforce</td> <td>Lead</td> </tr> </table>                                  | Opportunity field                   | Role | Skills & workforce      | Lead                                |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Opportunity field                                                                                                                                                                                        | Role                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |
| Skills & workforce                                                                                                                                                                                       | Lead                                |      |                         |                                     |                                                                                                                                                                                                                     |                   |      |                                                   |                                  |

## Conditional measures

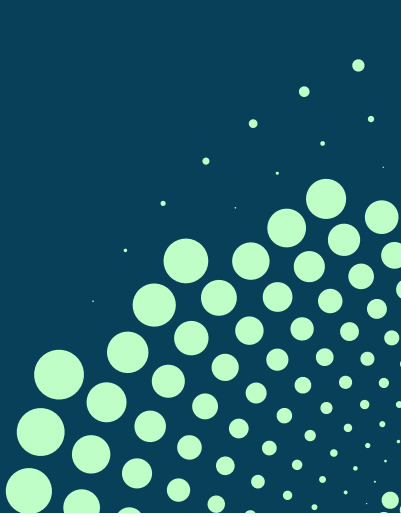
PC-1

Early warning system

|                   |         |
|-------------------|---------|
| Opportunity field | Role    |
| Policy handrails  | Support |



# Chapter 5: Update system: Research agent + expert panel + versioning



## 5.1 Purpose of this chapter

The AI Projections whitepaper is designed to be a living document, updated continuously as the AI landscape evolves. This chapter specifies the mechanisms through which we keep the document current, credible, and decision-relevant.

Core principle: The value of projection planning depends on timely updates when evidence shifts. A document that becomes outdated loses decision-relevance precisely when it matters most.

Invitation to participate:

We welcome contributions across all aspects of this document:

- Projections (Chapter 1): Perspectives on projection framing, baseline conditions, and projection descriptions
- Drivers & Probabilities (Chapter 2): Insights on main drivers, important trends, and probability assessments
- Impacts (Chapter 3): Observations on implications across impact fields; additional evidence or perspectives
- Measures (Chapter 4): Additions, ideas, and comments on proposed measures
- Collaboration (Chapter 5): If you would like to collaborate with us on our activities or if you want us to highlight your activities

Contact: [[ai-projections@appliedai-institute.de](mailto:ai-projections@appliedai-institute.de)]

## 5.2 Update triggers and cadence

### 5.2.1 Scheduled updates

| Update type        | Cadence         | Scope                                                                                                  | Output                                                      |
|--------------------|-----------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Annual edition     | Every 12 months | Full document revision; all chapters re-viewed and updated                                             | New version number (e.g., 2.0)                              |
| Quarterly re-view  | Every 3 months  | Benchmark updates; driver reassessment; probability check                                              | Version increment (e.g., 1.4)                               |
| Monthly monitoring | Every month     | Scan for all triggers including breakthrough triggers; flag emerging developments; update if warranted | Internal report; public update if material changes detected |

## 5.3 Research agent system

### 5.3.1 Continuous monitoring infrastructure

We maintain systematic monitoring across key information sources:

Primary sources:

| Source category                     | Examples                                                                                               | Monitoring frequency |
|-------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------|
| Benchmark leaderboards              | SWE-bench, GAIA, OSWorld, ARC-AGI-2, HumanEval, LiveCodeBench, METR, Vending-Bench 2                   | Weekly               |
| Major lab and company announcements | OpenAI, Anthropic, DeepMind, Meta AI, Mistral, xAI, AgiBot, UBTech, Unitree, Figure AI, Tesla Robotics | Daily                |
| Academic publications               | arXiv (cs.AI, cs.LG, cs.CL), NeurIPS, ICML, ICLR                                                       | Weekly               |
| Industry reports                    | McKinsey, BCG, Gartner, Stanford AI Index                                                              | As published         |
| Regulatory developments             | EU AI Office, national AI authorities, standards bodies                                                | Weekly               |

|                    |                                                             |           |
|--------------------|-------------------------------------------------------------|-----------|
| Market data        | AI investment tracking, startup announcements, M&A activity | Quarterly |
| Security incidents | CISA, ENISA, major incident reports                         | Daily     |

### 5.3.2 Signal processing

Raw monitoring data is processed through structured analysis:

1. Relevance filtering: Does this development affect one of our four drivers or baseline conditions?
2. Magnitude assessment: Is this incremental progress or a potential step-change?
3. Verification: Can the claim be corroborated by multiple independent sources?
4. Implication mapping: Which projections, impacts, or measures are affected?
5. Update recommendation: Does this warrant document revision?

### 5.3.3 AI-Assisted research

We use AI systems to support (not replace) human analysis:

- Literature scanning: Automated identification of relevant new publications
- Benchmark tracking: Automated extraction of performance updates from leaderboards
- Trend detection: Pattern identification across multiple information streams
- Draft preparation: Initial drafting of update sections for human review

Human oversight requirements:

- All AI-generated content is reviewed by human analysts before inclusion
- Probability estimates and strategic judgments are made by humans
- Expert panel validates significant updates before publication
- Sources are independently verified by human researchers

## 5.4 Expert panel

To be included in the future.

We plan to establish an expert panel representing diverse perspectives essential for comprehensive assessment, including domains such as AI research, AI engineering, AI safety, industry/enterprise, poli-

## 5.5 Versioning and change management

### 5.5.1 Version numbering

We use semantic versioning adapted for strategic documents:

| Version Format | Meaning                           | Example              |
|----------------|-----------------------------------|----------------------|
| X.0            | Major annual edition              | 2.0 (2027 edition)   |
| X.Y            | Quarterly or significant up-date  | 1.4 (Q2 2026 update) |
| X.Y.Z          | Minor correction or clarification | 1.3.1 (typo fix)     |

### 5.5.2 Change documentation

Every update includes:

- Version number: Clear identification of document version
- Publication date: When the update was released
- Change summary: High-level description of what changed
- Detailed changelog: Section-by-section list of modifications
- Rationale: Why changes were made (evidence basis)

### 5.5.3 Archive and access

- All previous versions are archived and accessible
- Current version is clearly marked as authoritative
- Comparison tools available to track changes between versions
- Subscription notification for stakeholders when updates published

## 5.6 Quality assurance

### 5.6.1 Accuracy standards

| Standard                   | Implementation                                             |
|----------------------------|------------------------------------------------------------|
| Source verification        | All factual claims traced to primary sources               |
| Benchmark accuracy         | Performance figures verified against official leaderboards |
| Expert attribution         | Quotes and positions verified with original sources        |
| Recency                    | Evidence dated; outdated information flagged for update    |
| Uncertainty acknowledgment | Confidence levels stated; speculation clearly marked       |

## 5.7 Contact and contribution

For updates, corrections, and collaboration inquiries:

- Email: [[ai-projections@appliedai-institute.de](mailto:ai-projections@appliedai-institute.de)]
- Web: [[appliedai-institute.de/ai-projections](https://appliedai-institute.de/ai-projections)]

We actively welcome:

- Feedback on projections, drivers, impacts, and measures
- New evidence and research findings
- Collaboration proposals on activities
- Requests to highlight relevant activities and initiatives

### Appendix A: Glossary of key terms

| Term       | Definition                                                                                                                                  |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Agentic AI | AI systems that plan and execute multi-step tasks using tools, interfaces, and memory, operating under varying degrees of human supervision |

|                                |                                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------|
| AI scare trade                 | Market sell-offs targeting sectors vulnerable to AI automation, triggered by new AI capability demonstrations   |
| ARC-AGI                        | Abstraction and Reasoning Corpus benchmark testing novel problem-solving ability                                |
| Automation frontier            | The boundary of tasks that can be reliably performed by AI systems with acceptable quality, cost, and oversight |
| Causal AI                      | AI approaches focused on understanding cause and effect rather than pattern correlation                         |
| Claude code                    | Anthropic's AI coding tool operating as an autonomous „AI Engineer“                                             |
| Few-person unicorn             | A billion-dollar company operated by a very small team (potentially single digits) enabled by AI automation     |
| Frontier models                | State-of-the-art general-purpose AI models defining current capability ceiling                                  |
| GAIA                           | General AI Assistants benchmark testing multi-step research and reasoning                                       |
| Nested learning                | DeepMind architecture treating models as interconnected multi-level optimisation problems                       |
| No-regret measure              | Action valuable regardless of which projection materializes                                                     |
| OpenClaw                       | Framework for system-integrated AI agents with local execution capabilities                                     |
| O-ring automation              | Pattern where AI generates candidates at scale but human filtering remains the bottleneck                       |
| OSWorld                        | Benchmark testing AI agents in real operating system environments                                               |
| Revenue per employee           | Key efficiency metric; AI-native companies achieving \$2-5M vs. \$400-500k traditional                          |
| Projection-conditional measure | Action that becomes critical primarily in specific projections                                                  |
| Sim-to-real transfer           | Process of translating AI capabilities trained in simulation to physical-world deployment                       |

|            |                                                                                   |
|------------|-----------------------------------------------------------------------------------|
| SWE-bench  | Software Engineering Benchmark using real GitHub issues to test coding capability |
| VLA models | Vision-Language-Action models enabling robots to perceive, understand, and act    |

#### Appendix B: Key benchmarks quick reference

| Benchmark          | Category             | Current SOTA | Human baseline | Status           |
|--------------------|----------------------|--------------|----------------|------------------|
| SWE-bench bash     | Software engineering | 76.8%        | ~70-80%        | Human parity     |
| SWE-bench Verified | Software engineering | 79.2%        | 100%           | Near-human       |
| GAIA               | Research/reasoning   | 74.55%       | 92%            | Closing gap      |
| OSWorld            | OS interaction       | 73%          | 72%            | Near-human       |
| WebArena           | Web tasks            | 57.1%        | ~85%           | Significant gap  |
| ARC-AGI            | Novel reasoning      | 75.7%        | ~85%           | Closing gap      |
| HumanEval          | Code generation      | 96.2%        | ~95%           | Saturated        |
| LiveCodeBench      | Fresh code problems  | 91.7%        | N/A            | High performance |

## Appendix C: Projection quick reference

| Dimension         | P1: Plateau (5%) | P2: Continued pace (50%) | P3: Accelerated (45%)       |
|-------------------|------------------|--------------------------|-----------------------------|
| Capability speed  | Marginal         | Steady                   | Compounding                 |
| Agentic viability | Limited          | Growing                  | Substantial autonomy        |
| R&D automation    | Assists          | Accelerates              | Feedback loops              |
| Robotics          | Gap persists     | Viable deployment        | Economically transformative |
| Europe's risk     | Complacency      | Competitiveness gap      | Loss of agency              |
| Urgency           | Moderate         | High                     | Very high                   |

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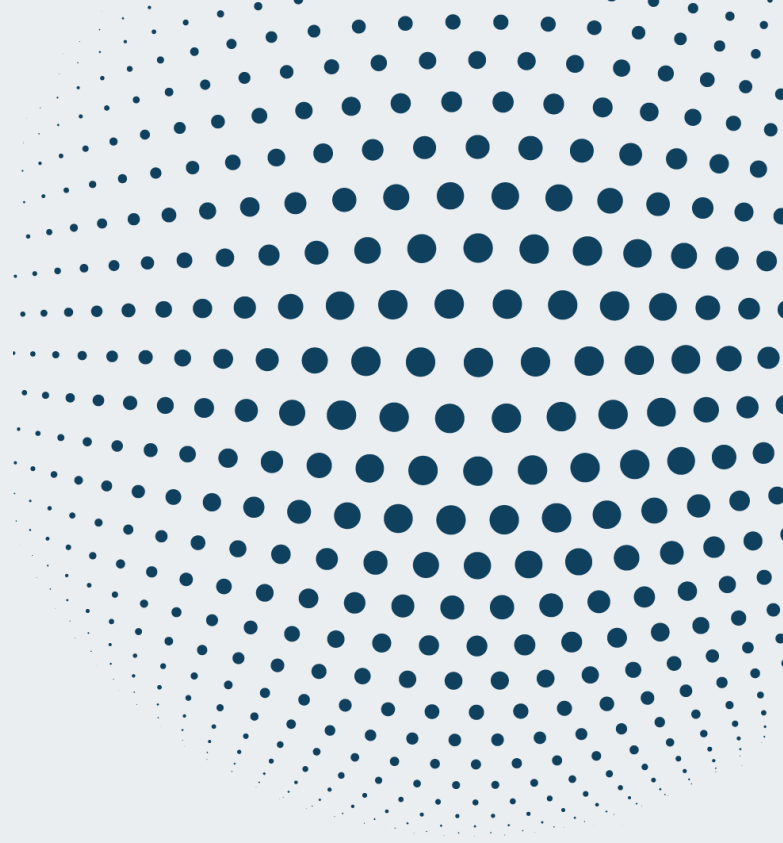
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